Supplementary methods. Details about statistical analysis

To clarify the effects of blood pressure lowering agents on the relative risk of stroke, ischemic stroke, hemorrhagic stroke, fatal or disabling stroke, cardiovascular death, and all cause death, we combined trials of blood pressure lowering agents versus placebo and higher versus lower blood pressure lowering targets and performed traditional meta–analysis. Two trials which compared angiotensin II receptor blocker (ARB)+angiotensin-converting enzyme inhibitor (ACEi) with ACEi were also included for the first objective.[1,2] For another two trials with two active groups and a placebo group,[3,4] we combined the events of the active groups for blood pressure lowering analysis. For one trial with three different blood pressure lowering targets,[5] we combined the two lower targets for analysis. We calculated relative risks from the number of events and participants for each outcome in each trial and pooled results with Mantel-Haenszel and Hartung-Knapp adjustment for random effects models. Random model other than fixed model was chosen because the included trials differed to some extent, both clinically and methodologically, and random model is generally more conservative compared with fixed effects model if heterogeneity is present. We assessed the magnitude of statistical heterogeneity among studies using standard cochrane chi-square test, the I^2 statistic (I^2values of at least 50% were considered to represent substantial heterogeneity, while values of at least 75% indicated considerable heterogeneity).[6] We explored evidence for heterogeneity in estimates of treatment effect attributable to the baseline characteristics of trials by comparing summary results obtained from subsets of studies grouped by age, history of cardiovascular disease, history of stroke, history of diabetes, baseline, and achieved blood pressure level. Publication bias was evaluated both graphically using a funnel plot and with the Egger statistical test for funnel plot asymmetry,[7] if at least 10 studies were available for each outcome. A leave-one-out sensitivity analysis was performed by repeating the meta-analysis, each time with one of the included studies omitted, to see whether any one study had disproportionately large impact on the pooled relative risk. Data used in this meta-analysis were intention to treat because most of the included trials did not report as treated results.

To clarify the efficacy and tolerability of different blood pressure lowering drugs for prevention of stroke, we combined the three groups of trials and did pair-wise and network meta-analysis. Ten trials of different blood pressure lowering targets which undefined any specific drugs were excluded from analysis. The outcome measure for efficacy and tolerability are stroke and drug-related side effects withdraw, respectively. First, we did pair-wise meta-analysis with a random effects model to analyze direct treatment comparisons. We calculated the summary effect sizes as odds ratios, with 95% confidence intervals. We assessed heterogeneity among studies with the I^2 statistic. We did not do funnel plots to test publication bias because most of the comparisons had less than 10 trials. Second, we analyzed pooled data for all blood pressure lowering treatments with random effects models, within a Bayesian framework in OpenBUGS.[8] See Appendix 2 for details about the OpenBUGS codes used. Models were computed with Markov chain Monte Carlo simulations, using three chains with over-dispersed initial values, with Gibbs sampling based on 100,000 iterations after a burn-in phase of 50,000 iterations. Non-informative or vague priors for the overall mean effect (θ to N (0, 1002)) and the between-study standard deviation (τ to uniform (0, 2)) were given.[9-11] The mean of the posterior distribution was reported as the point estimate odds ratio, and the corresponding 95% credible intervals were obtained with the 2.5th and 97.5th percentiles of the posterior distribution, after adjustment for multiple arm trials. We tested the adequacy of burn-in and convergence (reaching a stable equilibrium distribution) using visual inspection of parameter fluctuation depicted in trace plots and estimating the values of the Brooks-Gelman-Rubin statistic.[12] Model fit was evaluated with the total residual deviance, which indicated good fit, if it approximated the number of data points.

Inconsistency between direct and indirect evidence can suggest that the transitivity assumption might not hold. We assessed evidence for consistency in the networks in two ways. First, we used node-split approach to contrast direct evidence with indirect evidence from the entire network on each node.[11,13,14] A Bayesian *P*-value was calculated to estimate difference between direct and indirect evidence by counting the proportion of times the direct treatment effect exceeded the indirect treatment effect.[14] Second, we used the design-by-treatment interaction model that provides a single inference, using the chi-square test, about the plausibility of assuming consistency throughout the entire network.[15]

The surface under the cumulative ranking curve (SUCRA) and rankograms was used to provide a hierarchy of the regimens.[16] We also used two-dimensional plots and clustering methods to obtain meaningful groups of the treatments.[17] We assessed small study effects with comparison adjusted funnel plot symmetry.[17]

To investigate the generalisability of the findings, we assessed the effects of differing trial and participant characteristics on the outcomes in sensitivity analyses by restricting analyses to

studies with the following design characteristics: hypertensive participants; excluding heart failure participants; published in 2000 or later; duration of follow-up longer than 3 years. We did subgroup analyses according to age (age ≤60 and >60 years), history of stroke (no defines as participants with a history of stroke account for less than 5% of overall participants in a trial, yes defines as all of the participants has a history of stroke in a trial), history of diabetes mellitus (DM; no defines as participants with a history of DM account for less than 5% of overall participants in a trial, yes defines as all of the participants has a history of DM in a trial), and baseline systolic blood pressure (≤150 or >150 mm Hg).

For traditional meta-analyses we used R version 3.4.1 (R Foundation for Statistical Computing, Vienna, Austria). For network meta-analyses we used OpenBUGS 3.2.3 and STATA 14.0 (StataCorp., College Station, TX, USA).

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Supplementary Table 1. Baseline characteristics of included studies

Study	Intervention		Mean age (yr)		No. of female (%)		No. of participants		No. of participants with these conditions at baseline (%)				No. of participants with these conditions at baseline (%)				Follow-up duration	Mean SBP/DBP intervention group (mm Hg)		Mean SBP/DBP control group (mm Hg)		Drug-related side effect/withdrawal		Outcomes (intervention/control)			
	Inclusion criteria	Type of drug	Intervention	Control	Intervention	Control	Intervention	Control	CVD	CHD	Stroke	Hyper tension	PAD	DM	HF	AF	CKD		Baseline	Achieved BP	Baseline	Achieved BP	Inter-vention	Control	all-cause death	Death from all causes	Cardio-vascular death
Action[14]	Stable angina+ history of MI/CAD	CCB (nifedipine GITS)	63.5±9.3	63.4±9.3	744 (39.7)	797 (42.5)	3,825	3,840	7,665 (100)	7,665 (100)	NA	3,977 (51.9)	985 (12.9)	1,170 (15.3)	0	NA	4.9	137.3/78.8/ 79.9±9.4	130.3/75.8	137.6/ 78.6/ 79.8±9.5	NA	389	172	827/08	310/291	178/177
Active I[15]	AF+risk factors for stroke	ARB (candesartan)	69.5±9.7	69.6±9.7	1,273 (39.2)	1,768 (39.3)	4,568	4,498	9,016 (100)	NA	1,212 (13.4)	7,929 (87.9)	236 (2.6)	1,797 (19.8)	2,081 (23.1)	9,016 (100)	NA	4.1	138.3/-/77.6/ 82.6±11.5	138.2/-/77.2/ 82.2±11.1	NA	NA	NA	379/411	949/929	NA
ADVANCE[16]	DM+at least 1 risk factor for CVD	ACEI (perindopril) ± diuretic (indapamide)	66±6	66±7	2,366 (29.6)	2,369 (29.6)	5,569	5,571	NA	1,334 (23.9)	1,022 (18.4)	NA	11,140 (100)	NA	NA	401 (4.0)	4.3	145±22/ 81±11.5	136/73	145±21/ 81±11.5	NA	300	125	215/218	408/471	211/257
Attitude[17]	DM+≥1 microvascular/ macrovascular complication/ afvd/CVD	Renin inhibitor (aliskiren)	64.6±9.6	64.4±9.3	1,393 (30.1)	1,342 (29.1)	4,274	4,287	3,619 (84.7)	NA	847 (19.8)	8,986 (84.0)	NA	8,561 (100)	872 (20.4)	731 (17.1)	8,990 (84.0)	2.7	137.3±16.2/ 78.1±9.8	139/75	137.3±16.1/ 78.1±9.8	NA	325	221	1417/1372	376/358	246/215
AIRE[18]	AMI+HF/EF<40%	ACEI (ramipril)	64.9±10.8	65.1±10.8	270 (29.7)	255 (28.0)	1,004	982	1,986 (100)	1,986 (100)	NA	554 (28)	NA	240 (12)	1,986 (100)	NA	0	1.3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Australian trial[19]	Mild HTN without risk for CVD	ARB (candesartan)	50.4±9.0	50.5±8.9	804 (37.0)	850 (36.4)	1,721	1,706	NA	NA	NA	NA	NA	NA	NA	NA	NA	4	137.7±15.0/ 100.5±9.0	NA/88.3	137.1±14.4/ 100.4±8.8	NA/91.9	NA	NA	10/16	25/35	8/18
BCAPS[20]	Plaque in the right carotid artery+no symptoms of CAD	BB (metoprolol CR/XL)	61.6±5.4	61.9±5.3	221 (22)	211 (21)	396	397	34 (4.3)	NA	NA	NA	NA	95 (12.1)	25 (3.2)	NA	NA	3.1	138.5/ 84.7	NA	139.4/ 84.8	NA	79	87	1/7	4/7	NA
BHAT[21]	Hospitalized for MI	Propranolol	NA	54.7	54.9	310 (17.0)	286 (15.7)	1,916	1,921	3,837 (100)	3,837 (100)	NA	1,664 (43.4)	NA	441 (11.5)	353 (9.2)	NA	2.1	113.37/72.5	127/80	111.77/72.3	130/81	243	179	29/30	138/188	127/171
CAMELOT[22]	CAD 1-20% stenosis by coronary angiography+DBP <100 mm Hg	ACEI (enalapril)	57.3±9.7	57.2±9.5	157 (23.7)	177 (27)	663	655	1,318 (100)	1,318 (100)	51 (3.9)	802 (60.8)	NA	245 (18.6)	0	NA	NA	2	129.5±15.5/ 77.7±9.1	124.7/75.2	128.9±15.8/ 77.6±9.9	NA	NA	NA	67/2	7/6	5/2
DIABHYCAR[23]	DM+microalbuminuria or proteinuria+serum creatinine <150 mol/L	ARB (candesartan)	65.2±6.4	65.0±6.3	742 (34.0)	738 (29.9)	2,443	2,469	1,201 (49.2)	295 (12.1)	207 (8.5)	2,735 (42.2)	503 (25.7)	4,912 (10.2)	390 (100)	4.9	145.8± 15.0/ 82.4 ±8.7	145.1±15.2/ 82.2±8.5	609	554	2017/200	NA	3200/308			
DIRECT-Protect 2[24]	DM+normoalbuminuric, normotensive, or treated hypertensive+mild to moderately severe retinopathy	ARB (candesartan)	56.9±7.6	56.8±7.9	485 (51.0)	472 (49)	951	954	125 (13.1)	99 (10.4)	26 (2.7)	1,180 (1.4)	NA	1,905 (100)	NA	NA	4.7	123.9/ 75±6 (normotensive) 129.7/ 79±7 (hypertensive)	128/74	125.9/78.9±9	132/76	NA	NA	623	NA	62/4	
Dream[25]	Non-diabetic+impaired fasting glucose level+at impaired glucose tolerance+no CVD	ACEI (ramipril)	Placebo	54.7 ±9.9	54.7 ±10.1	1,567 (59.7)	1,553 (58.7)	2,623	2,646	0	0	0	2,291 (87.3)	0	0	0	0	3	136.1±16.6/ 80.4±10.3	127/87/8	136.0±18.1/ 83.4±10.8	132.1/ 80.4	798	615	4/8	31/32	12/10
Dutch TIA[26]	TIA/nondisabling cerebrovascular stroke less than 3 months before	BB (atenolol)	NA	NA	NA	249 (23.5)	282 (26.7)	732	741	NA	81 (11.0)	1,473 (100)	420 (57.1)	37 (5.0)	74 (10.1)	NA	NA	2.6	158±25/ 91±13.5	149.5/NA	157±24/ 91±13.5	NA	NA	NA	52/62	64/58	41/33
EUROPA[27]	Stable CAD without HF	ACEI (perindopril)	60±9	60±9	884 (14.5)	895 (14.7)	6,110	6,108	12,218 (100)	12,218 (100)	409 (3.3)	3,512 (27.1)	883 (7.2)	799 (6.5)	0	NA	4.2	137 (16)/ 82 (8)	NA	137 (15)/ 82 (8)	NA	1,391	1,266	98/102	376/420	215/249
EWPHE[28]	HTN DBP 90-119 mm Hg and SBP 160-239 mm Hg untreated	Diuretic (chlorothiazide)+K-sparing diuretic (triamterene)	72±8	72±8	287 (59)	287 (59.8)	416	424	NA	NA	NA	NA	NA	840 (100)	NA	NA	4.7	164±12/ 101±7	148+18/ 85±10	165±12/ 101±7	167±22/ 90±9	NA	NA	12/19	73/89	42/61
FEVER[29]	Untreated HTN: DBP ≥115 mm Hg and SBP ≤210 mm Hg Treated HTN: SBP 160–210 mm Hg or DBP 95–115 mm Hg	CCB (felodipine)+diuretic (hydrochlorothiazide)	61.5 ±7.1	61.5 ± 7.2	1,858 (38.2)	1,833 (38.0)	4,841	4,870	NA	1,318 (14.8)	1,438 (16.1)	9,711 (100)	48 (0.5)	1,241 (13.9)	614 (6.9)	NA	3.3	158.7 ±17.6/ 92.4 ± 9.6	138.1±11.4/ 82.2± 7.3	158.9±17.3/ 92.7±9.6	141.6±12.2/ 82.3±7.7	NA	NA	1772/551	1121/551	737/301
Felodipine[30]	Physician's choice	CCB (felodipine)	Placebo	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FOSTER[31]	DM+HF (NYHA II-IV)+LVEF ≥40%+≥60 years	ARB (telmisartan)	72±7	72±7	2,050 (26.5)	2,031 (27.2)	8,502	8,576	NA	2,067 (24.3)	2,060 (24.2)	NA	7,469 (87.8)	3,309 (38.9)	3,650 (42.9)	NA	4.2	128.4 ±16.2/ 77.2 ±9.2	128.1 ±16.2/ 77.2 ±9.2	128.4 ±16.2/ 77.2 ±9.2	128.1 ±16.2/ 77.2 ±9.2	NA	NA	1,391	1,266	98/102	376/420	215/249
ONTARGET[32]	Vascular disease or DM+high risk	ARB (telmisartan)+ ACEI (ramipril) ARB (telmisartan)	66.6±7.3	66.4±7.2	2,250 (26.5)	2,331 (27.2)	8,502	8,576	NA	NA	NA	NA	NA	7,464 (87.8)	2,307 (68.8)	1,116 (13.5)	0.4	1,858 (11)	4.7	137.7/76.1/ 82.1 ±10.4	NA	137.6/76.1/ 82.1 ±10.4	NA	NA	NA	3734/05	1,065/ 1,014	620/693
OSCAR[33]	HTN aged 65–84 years+DM/CVD	ARB (olmesartan)+CCB (amlodipine or azelnidipine)	73.6±5.5	73.6±5.5	325 (55.0)	324 (55.0)	586	578	812 (100)	NA	207 (25.5)	1,164 (100)	25 (3.1)	NA	89 (11.0)	NA	127 (10.8)	3	157.2±11.2/ 84.6±8.8	132.6/72.6	158.2±12.6/ 84.9±9.2	135.0/74.3	NA	NA	15/24	NA	NA
Oslo Study[34]	HTN aged 20–49 years DBP 150–179 mm Hg, DBP <110 mm Hg	Diuretics (hydrochlorothiazide)/ß-adrenoreceptor agonist (bupranolol)/ peripheral vasodilator (dihydralazine)	45.3±2.9	45±2.8	0	0	406	379	0	0	0	785 (100)	0	0	0	0	5.5	156.2±7.1/ 97.4±6.9	0	155.3±7.6/ 96.2±7.2	0	NA	NA	0/5	NA	NA
PART-2[35]	CHD/CVD without HF	Diuretic (indapamide)	60±8	61±8	56 (17.9)	56 (18.1)	308	309	617 (100)	259 (42.0)	20 (3.2)	617 (100)	40 (6.5)	38 (6.2)	0	NA	4.7	123±17/ 79±11	127/74	123±16/ 79±10	132/78	31	3	7/4	16/25	8/18
PATS[36]	History of TIA/ non-disabling stroke	Diuretic (indapamide)	60.1±8.3	60.4±8.5	803 (40.1)	785 (39.2)	2,840	2,825	NA	NA	5,665 (100)	4,752 (83.9)	NA	NA	0	0	2	153.6±23.8/ 82.6±13.1		154.0±23.3/ 82.7±13.1						
PEACE[37]	CAD	ACEI (trandolapril)	Placebo	64±8	64±8	790 (17.0)	702 (15.1)	4,158	4,132	8,290 (100)	8,290 (100)	539 (6.5)	3,772 (45.5)	NA	1,410 (17.0)	0	NA	4.8	134/ 77	133/ 75	135/ 79	133/ 77	599	269	71/92	299/334	149/152
PHARAO[38]	Aged 50–85 years with high-normal office BP	ACEI (ramipril)	Placebo	62.2±8.2	62.2±7.9	254 (30.3)	266 (32.0)	505	503	NA	65 (12.9)	NA	NA	1,008 (100)	15 (1.5)	135 (13.4)	0	3	134.4±3.7/ 83.6±4.2	130.2/79.0	134.4±3.4/ 83.6±4.7	133.0/79.9	NA	NA	2/1	5/2	0/0
PREVEND IT[39]	Microalbuminuria	ACEI (fosinopril)	51.1±12.2	51.5±11.4	1,174 (30.9)	1,152 (30.3)	4,237	3,810	431 (10.2)	433 (11.4)	29 (0.7)	0	7 (0.2)	0	0	0	3.8	129±13.7/ 76±10	129±18	131±14.2/ 76±10	132±18	58	18	1/10	NA	5/3
PREVENT[40]	CAD	CCB (amlodipine)	56.8	57	84 (20.1)	89 (21.4)	417	408	0	825 (100)	0	0	0	0	0	NA	3	128.8/ 80.2±8.6	122/75	128.8/ 80.2±8.6	122/75	NA	NA	5/5	6/8	NA
PRoFESS[41]	Ischaemic stroke	ARB (telmisartan)	66.1±8.6	66.2±8.6	2,503 (35.7)	2,502 (36.2)	10,146	10,186	NA	NA	NA	20,332 (100)	1,048 (5.2)	5,743 (28.2)	540 (2.7)	NA	2.5	144.1±16.4/ 83.8±10.5	135/79	144.2±16.7/ 83.8±10.6	NA	1,450	1,127	880/934	759/740	223/262
PROGRESS[42]	Stroke/TIA+HTN	ACEI (perindopril)± diuretic (indapamide)	64±10	64±10	923 (30.2)	1,003 (32.8)	3,051	3,054	6,105 (100)	983 (16.1)	6,105 (100)	1,948 (31.9)	244 (4.0)	1,148 (18.8)	0	488 (8.0)	3.9	147/82/ 86±11	134/79	148/ 86±11	144/ 85	111	98	307/420	306/319	181/198
ROADMAP[43]	DM+2 risk factors+albuminuria	ARB (olmesartan)	57.7±8.8	57.8±8.6	1,182 (53.0)	1,212 (54.7)	2,232	2,215	NA	1,594 (71.4)	104 (4.7)	NA	NA	NA	25 (1.1)	4,447 (99.4)	NA	0.2	127±16/ 81±10	125.7/74.3	126±15.8/ 80±9	128.7/76.2	85	89	NA	26/15	15/3
SCOPE[44]	HTN (mild-moderate)	CCB (candesartan)	76.4	76.4	1,905 (39.3)	1,779 (37.3)	2,477	2,460	NA	193 (7.8)	4,937 (100)	607 (24.5)	NA	NA	3.7	166/ 90.3	145.2/79.9	166.5/90.4	148.5/81.6	372	418	89/115	259/266	145/152	
SHEP[45]	Elderly ISH DBP 160–219 mm Hg, DBP <90 mm Hg	Diuretic (chlorthalidone), ß-blocker (atenolol)	71.6±6.7	71.5±6.7	1,339 (30.1)	1,361 (30.6)	2,365	2,371	NA	NA	66 (2.8)	4,736 (100)	NA	478 (10.1)	NA	NA	4.5	170.5±9.5/ 76.1±5.4	144.0±19.3/ 74.3±8.9	170.1±9.2/ 76.1±5.4	155.1±20.8/ 71.1±7.8	307	166	967/49	213/242	90/112
SOLVD-Treatment[46]	CHF+EF ≤35%	ACEI (enalapril)	59.1	59.1	241 (20.1)	257 (21.4)	2,111	2,117	4,228 (100)	1,228 (58.2)	1,228 (58.2)	1,228 (58.2)	1,228 (58.2)	1,228 (58.2)	4,228 (100)	167 (7.9)	NA	3.1	125.3/ 77.9	NA	126.4/78.0	NA	NA	NA	1013	313/334	265/298
SUPPORT[47]	CHF+HTN	ARB (irbesartan)	65.8±9.4	65.6±9.5	141 (25.8)	147 (24.8)	578	569	545 (94.3)	NA	NA	NA	NA	575 (99.5)	1,147 (100)	NA	NA	4.4	128.7±18.2/ 74.8±12.2	NA	127.1±18.0/ 73.9±11.7	NA	NA	NA	34/26	98/86	48/58
STONE[48]	Elderly HTN SBP 160–79 years, SBP≥ 180 mm Hg, DBP ≥90 mm Hg for DBP 105–120 mm+period of SBP)	CCB (nifedipine)	66.1	66.7	408 (43.9)	409 (43.9)	817	815	NA	NA	NA	1,632 (100)	NA	NA	NA	NA	2.5	158.1±18.2/ 98.5±7	NA	158.5±19.2/ 97±7	NA	NA	NA	16/36	15/26	11/14
STOP-Hypertension-2[49]	388+1 diuretic (atenolol, hydrochlorothiazide, amlodipine, metoprolol, enalapril, lisinopril)	CCB (nitrendipine),ACEI (captopril) and/or diuretic (hydrochlorothiazide)	76.6±3.7	75.7±3.7	510 (56.1)	509 (56.0)	812	815	NA	NA	NA	1,627 (100)	NA	NA	NA	NA	2.1	195±14/ 102±7	166±21/ 85±10	195±14/ 102±7	193±20/ 95±11	NA	NA	29/53	36/63	17/41
Syst-China[50]	Elderly (≥60 years, SBP 160–219 mm Hg, DBP <95 mm Hg)	CCB (nitrendipine),ACEI (captopril) and/or diuretic (hydrochlorothiazide)	66.4±5.4	66.7±5.7	438 (55.0)	415 (54.6)	1,253	1,141	269 (21.5)	224 (1.4)	34 (2.7)	2,394 (100)	98 (4.1)	NA	NA	NA	3	170.7±10.9/ 86.1±6.7		170.2±11.4/ 85.9±7.0		NA	NA	49/59	61/82	33/44
Syst-Eur[51]	ISH elderly (≥60 years, SBP 160–219 mm Hg, DBP <95 mm Hg)	CCB (nitrendipine),ACEI (enalapril) and/or diuretic (hydrochlorothiazide)	70.3±6.7	70.2±6.7	1,520 (61.8)	1,618 (65.8)	2,398	2,297	1,402 (58.5)	NA	NA	4,695 (100)	NA	NA	NA	NA	2	173.8±9.9/ 75.9±10.6		173.9±10.1/ 85.5±9.9		NA	NA	47/77	123/137	59/77
TRIAN-SCEND[52]	ACEI resistance+CVD/ DM	ACEI (ramipril)+ARB (telmisartan)	66.9±7.3	66.6±7.4	1,280 (43.3)	1,267 (42.6)	2,954	2,972	NA	4,418 (42.6)	1,302 (12.6)	4,528 (76.4)	672 (11.3)	2,118 (35.7)	0	NA	4.7	140.7±16.6/ 81.8±10.1	134.2/NA	141.3±16.4/ 82.0±10.2	NA	NA	NA	112/136	364/349	227/223
VA-HFH-RON[53]	AMI+symptomatic	ACEI (lisinopril)+ARB (losartan)	64.5±7.9	64.7±7.7	Placebo (1.2)	ARB (losartan) (1.4)	724	724	NA	NA	NA	NA	NA	NA	NA	NA	1.4	122.4/ 72.5±10.6	122/NA	122.4/ 72.8±9.3	NA	NA	NA	18/18	63/60	NA
AASK[54]	African American hypertensive CKD (DBP >95 mm Hg), without DM/HF	Diuretic (DBP/DBP <130/80 mm Hg)	54.5±10.9	54.7±10.4	204 (38.1)	206 (39.5)	542	504	564 (52.1)	NA	NA	1,084 (100)	NA	0	NA	0	3	150.4±21.5/ 96±15	128±21/ 78±14	150.4±21.5/ 96±15	140±18/ 86±11	NA	NA	26/29	38/47	16/15
ALLHAT[55]	Elderly HTN (≥60 years)	BB (metoprolol)	54.4±9.0	54.9±9.4	168 (38.5)	170 (38.6)	436	441	NA	NA	NA	877 (100)	NA	0	NA	0	4.1	151±23/ 86±10	135±14/ 79±9	150±24/ 86±10	135±12/ 81±9	NA	NA	23/23	NA	12/11
	BB	CCB (amlodipine)	54.5±9.7	54.9±9.4	170 (39.4)	170 (38.6)	217	441	NA	NA	NA	658 (100)	NA	0	NA	0	4.1	150±23/ 86±10	133±12/ 79±9	150±24/ 86±11	134±12/ 81±9	NA	NA	9/23	NA	7/12
	BB	ACEI (lisinopril)	54.4±9.3	54.9±9.4	168 (38.9)	166 (37.7)	436	217	NA	NA	NA	653 (100)	NA	0	NA	0	4.1	151±23/ 86±10	135±14/ 79±9	150±29/ 86±10	133±12/ 81±9	NA	NA	23/9	NA	12/7
ABCD-H[56]	DM+HTN	CCB (amlodipine)	57.3±5.5	57.2±5.5	Moderate DBP 90–99 mm Hg	Moderate DBP 90–99 mm Hg	75	237	233	NA	115 (15.4)	5 (1.1)	470 (100)	470 (100)	3 (0.6)	37 (7.9)	5	156.1±1.1/ 84.0±0.4	133/78	154.9±1.2/ 84.0±0.4	NA	NA	NA	9/9	13/25	NA
	DBP ≥90 mm Hg	Nisoldipine	57.2±8.2	57.7±8.4	75 (32.2)	78 (33.2)	235	235	NA	115 (12.2)	5 (1.2)	470 (100)	NA	470 (100)	3 (0.6)	37 (7.9)	5	155±19/ 98±7	NA	156±17/ 98±7	NA	NA	NA	9/11	17/13	10/5
ABCD-N[57]	Intensive (DBP 90 mm Hg below the baseline DBP)	Moderate (DBP 80–89 mm Hg)	58.5±9.6	59.6±9.5	111 (46.8)	107 (44.6)	237	243	NA	115 (12.2)	17 (1.8)	480 (100)	10 (2.1)	522 (100)	5 (1.0)	43 (9.2)	5	129.6±0.8/ 84.4±0.2	129±0.8/ 75±0.3	129.6±0.8/ 84.4±0.2	137±0.5/ 81±0.3	NA	NA	4/13	18/20	13/9
ACCORD[58]	AMI+high risk	Intensive SBP <120 mm Hg	62.2±6.9	62.2±6.9	Standard (SBP <140 mm Hg)		1,128 (24.1)	1,130 (24.1)	2,362	2,371	1,503 (32.1)	NA	NA	NA	NA	NA	4	4,733 (100)	203 (4.3)	NA	119.3/66.4 (100)	119.4±15.5/ 75.9±10.6	133.5/ 70.5	NA	NA	3682	1507/44	610/58
BBB[59]	Treated HTN DBP 90–100 mm Hg) without CHD	Usual (DBP 80–100 mm Hg)	59.6	60.1	NA	NA	1,064	1,063	NA	0	NA	2,127 (100)	NA	NA	NA	NA	4.9	155±15/ 91±10	141±15/ 84±9	154±15/ 91±10	152±15/ 91	NA	NA	87/1	NA	NA
Cardio-sis[60]	HTN SBP ≥ 150 mm Hg+1–3 risk factors	Intensive (SBP <130 mm Hg)	67±7	67±7	329 (52.4)	324 (52.2)	558	553	NA	128 (23.0)	91 (16.3)	1,111 (100)	3,020 (54.4)	317 (57.0)	3,020 (54.4)	1,106 (19.9)	NA	3.7	163.3±11.3/ 89.6±8.8		163.3±11.1/ 89.6±8.8		NA	NA	4/9	4/5	NA
HDFP[61]	HTN	Intensive (stepped care)	50.8	50.8	2,329 (51.1)	2,509 (54.1)	5,485	5,455	NA	564 (10.3)	273 (5.0)	10,940 (100)	NA	771 (14.1)	NA	NA	5	159.0/101.1	NA/84.1	158.5/101.1	NA/88.1	NA	NA	1002/158	349/419	199/240
HOMED-BP[62]	HTN+>40 years	Tight (125/80 mm Hg)	Usual DBP 125–134 mm Hg and DBP 80–84 mm Hg)	59.6±10.2	59.6±9.9	883 (50)	880 (50)	1,759	1,759	106 (6)	NA	NA	3,518 (100)	538 (15)	NA	NA	NA	5.3	151.5±12.3/ 90.0±9.8		151.7±12.6/ 89.9±10.3		NA	NA	20/16	27/31	3/5
		ACEI	ARB	59.8±10	59.9±10.1	589 (33.5)	588 (33.4)	1,172	1,175	75 (4.3)	NA	NA	2,347 (100)	372 (21.2)	NA	NA	5.3	151.6±12.9/ 89.8±10.5	129.3±13.3/ 80.3±8.1	151.6±12.4/ 89.8±10.4	129.8±13.0/ 80.5±8.1	NA	NA	11/9	17/16	2/2
		CCB	ACEI	59.5±10.1	59.8±10.1	586 (33.3)	588 (33.4)	1,171	1,172	65 (3.7)	NA	NA	2,343 (100)	347 (19.8)	NA	NA	5.3	151.6±12.9/ 90.1±9.9	130.1±13.3/ 80.9±8.2	151.6±12.4/ 89.9±10.4	129.3±13.3/ 80.5±8.2	NA	NA	16/21	25/17	4/2
		CCB	ARB	59.5±10.1	59.8±10.1	586 (33.3)	588 (33.4)	1,171	1,175	72 (4.1)	NA	NA	2,346 (100)	357 (20.3)	NA	NA	5.3	151.6±12.9/ 90.1±9.9	130.1±13.3/ 80.9±8.2	151.6±12.4/ 89.9±10.4	129.8±13.0/ 80.5±8.2	NA	NA	16/9	25/16	4/2
HOT[63]	DM+HTN	DBP 85 mm Hg	DBP 85 mm Hg	61.5±7.5	61.5±7.5	2,944 (31.9)	2,943 (31.9)	6,264	6,264	NA	745 (11.9)	150 (2.4)	12,528 (100)	1,002 (8.0)	NA	NA	3.8	139.4±14.1/ 89±9.2	141.4±11.7/ 89±9.2	139.4±14.1/ 89±9.2	143.2±11.2/ 91.2±9.3	NA	NA	111/94	194/188	90/87
		DBP 80 mm Hg	DBP 80 mm Hg	61.5±7.5	61.5±7.5	2,944 (31.9)	2,943 (31.9)	6,262	6,264	NA	739 (11.8)	150 (2.4)	12,526 (100)	1,000 (8.0)	NA	NA	3.8	139.4±14.1/ 89±9.2	139.7±11.7/ 89±9.2	139.4±14.1/ 89±9.2	143.2±11.2/ 91.2±9.3	NA	NA	111/94	207/188	96/87
		DBP 80 mm Hg	DBP 85 mm Hg	61.5±7.5	61.5±7.5	2,944 (31.9)	2,943 (31.9)	6,262	6,264	NA	745 (11.9)	150 (2.4)	12,526 (100)	1,000 (8.0)	NA	NA	3.8	139.4±14.1/ 89±9.2	139.7±11.7/ 89±9.2	139.4±14.1/ 89±9.2	143.4±11.7/ 91.4±9.4	NA	NA	89/111	207/194	96/90
JATOS[64]	Elderly+HTN	Strict (SBP <140 mm Hg)	Mild (SBP 140–159 mm Hg)	73.6±5.3	73.6±5.2	1,338 (61.2)	1,363 (62.3)	2,212	2,206	NA	134 (6.1)	192 (8.8)	4,418 (100)	521 (23.8)	NA	NA	3	439 (20.1)	2	171.6±9.7/ 84.7±8.6	139.7±11.7/ 74.8±7.1	171.5±9.8/ 84.8±8.6	145.6±11.1/ 75.1±9.9	NA	NA	43/38	64/42	9/8
SPRINT[65]	High risk of CVD, >50 years, SBP 130–180 mm Hg, without DM2	Intensive (SBP <120 mm Hg)	67.9±9.4	67.9±9.5	1,684 (36.0)	1,648 (35.2)	4,678	4,683	1,877 (20.1)	NA	NA	9,361 (100)	NA	0	NA	0	3.3	1,397.7±15.8/ 78.2±11.9	121.5/NA	1,394.5±15.4/ 78.0±12.0	134.6/NA	NA	NA	63/70	155/210	37/65
SPS3[66]	Previous stroke	SBP <130 mm Hg	63±10.7	63±10.8	589 (53.6)	529 (48.1)	1,519	1,517	3,020 (100)	317 (10.5)	3,020 (100)	2,264 (75.0)	1,106 (36.6)	NA	NA	377 (12.5)	3.7	142.4±18.5/ 77.4±9.5	136.4±15.9/ 76.4±9.1	142.4±18.5/ 77.4±9.5	137.4±16.2/ 76.1±9.1	NA	NA	125/152	106/101	36/41
UKPDS 38[67]	HTN+DM2	Strict BP control (<150/85 mm Hg)	56.4±8.1	56.5±8.1	348 (40)	163 (42)	758	390	NA	NA	NA	1,148 (100)	NA	NA	NA	NA	8.4	159±20/ 94±10	144±14/ 82±7	160±18/ 94±10	154±16/ 87±7	NA	NA	38/34	134/83	NA
VALISH[68]	Aged ≥70 and <85 years with ISH (SBP >160 mm Hg and DBP <90 mm Hg)	Strict (SBP <140 mm Hg)	Moderate (SBP 140–159 mm Hg)	76.1±4.1	76.1±4.1	963 (62.3)	961 (62.5)	1,545	1,534	NA	153 (9.9)	202 (13.1)	3,079 (100)	399 (25.9)	53 (3.4)	NA	4.3	143 (2.9)	2.9	169.5±7.9/ 81.7±6.6	136.6±13.3/ 74.8±8.8	169.6±7.8/ 81.2±6.8	142±12.5/ 76.5±8.9	NA	NA	14/23	24/20	11/11
CAPPP[69]	DBP ≥100 mm Hg	Diuretic/BB	ACEI (captopril)	55.7±7.4	55.0±7.6	105 (6.1)	113 (6.5)	263	309	52 (3.0)	NA	NA	10 (0.6)	572 (33.1)	NA	572 (33.1)	9	57 (3.3)	6.1	163.3±20.6/ 97.3±10.1	153.5/88.0	163.6±18.6/ 97.1±10.1	155.5/ 89.0	NA	NA	19/23	NA	NA
IPPPSH[70]	HTN (DBP 100–125 mm Hg)	Diuretic/BB	Non-BB	52.3±6.5	52.2±6.5	1,583 (52.5)	1,583 (52.5)	3,185	3,172	0	0	0	6,357 (100)	0	NA	0	4	173/107.9	143.6/88.9	172/ 107.6	147.4/ 90.1	NA	NA	49/46	109/114	NA
NORDIL[71]	HTN DBP ≥100 mm Hg	Diuretic/BB	CCB (diltiazem)	60.3±6.5	60.5±6.5	2,805 (52.1)	2,786 (51.6)	5,471	5,410	NA	162 (3.0)	10,881 (100)	727 (13.5)	NA	101 (1.9)	NA	4.5	173.4±17.5/ 98.2±6.3	151.7/88.7	173.5±17.7/ 98.2±6.4	154.9/ 88.6	NA	NA	196/159	228/231	115/131
ACTOMPS[72]	HTN+high risk	CCB+ACEI (benazepril+amlodipine)	68.4±6.6	68.3±6.6	2,296 (43)	2,246 (39)	5,744	5,762	NA	2,709 (100)	1,498 (26.0)	11,506 (100)	6,946 (60.4)	0	769 (6.7)	705 (6.1)	3	145.3±18.4/ 80.1±10.8	131.6/73.3	145.4±19.1/ 80.0±10.7	132.5/ 74.4	1,654	1,798	112/123	230/262	107/134
ALLHAT[73]	HTN+CVD/risk factors for CVD	CCB (amlodipine)	Diuretic (chlorthalidone)	66.9±7.7	66.9±7.7	6,280 (47.3)	6,211 (47.0)	9,048	15,255	18,240 (75.1)	6,145 (25.3)	NA	24,303 (100)	NA	8,851 (36.4)	NA	NA	4.9	146±16/ 84±10	134.7±14.8/ 74.6±9	146±16/ 84±10	133.9±15.2/ 75.4±9.8	NA	NA	3778/75	1,256/ 2,203	592/992
		ACEI (lisinopril)	Diuretic (chlorthalidone)	66.9±7.7	66.9±7.7	6,213 (47.4)	6,117 (46.2)	7,477	7,171	9,004 (75.4)	6,213 (25.0)	NA	24,009 (100)	NA	8,740 (36.4)	NA	NA	4.9	146±16/ 84±10	135.9±13.7/ 75.4±10.7	146±16/ 84±10	133.9±15.2/ 75.4±9.8	NA	NA	4577/75	1,314/ 2,

Supplementary Table 2. Pairwise meta-analysis results of efficacy and tolerability for direct comparisons of interventions

Comparisons		Pairwise meta-analysis odds ratio (95% CI)	*P*	No. of trials	No. of participants	Heterogeneity I^2
Efficacy						
CCB	vs. Placebo	0.642 (0.539–0.765)	<0.001	7	19,665	0
ARB		0.919 (0.864–0.978)	0.008	13	69,891	0
ACEi		0.813 (0.747–0.885)	<0.001	13	58,691	49.4
Diuretic		0.632 (0.556–0.718)	<0.001	8	35,543	30.6
BB		0.791 (0.673–0.929)	0.004	7	29,667	20.8
ACEi+Diuretic		0.986 (0.814–1.195)	0.886	1	11,140	NA
CCB+Diuretic		0.698 (0.574–0.850)	<0.001	1	9,711	NA
ARB+Diuretic		0.795 (0.585–1.078)	0.140	1	12,705	NA
Diuretic/BB		0.532 (0.335–0.847)	0.008	1	1,627	NA
Renin inhibitor		1.209 (0.954–1.531)	0.116	1	8,561	NA
ARB	vs. CCB	0.977 (0.869–1.099)	0.698	8	29,011	72.3
ACEi		1.123 (1.008–1.252)	0.036	8	29,533	12
Diuretic		1.063 (0.947–1.194)	0.300	4	33,920	0
BB		1.261 (1.116–1.425)	<0.001	4	4,825	0
Diuretic/BB		0.922 (0.762–1.114)	0.399	3	31,766	55.8
ACEi	vs. ARB	1.074 (0.972–1.186)	0.159	5	37,912	5.5
BB		1.361 (1.142–1.622)	0.001	1	9,193	NA
ARB+ACEi		1.017 (0.908–1.139)	0.771	3	28,320	0
ARB+CCB		0.606 (0.315–1.168)	0.135	1	1,164	NA
Diuretic	vs. ACEi	0.871 (0.771–0.984)	0.026	1	24,309	NA
BB		0.988 (0.546–1.789)	0.968	1	877	NA
Diuretic/BB		1.097 (0.911–1.321)	0.329	2	4,990	0
ARB+ACEi		0.909 (0.813–1.017)	0.097	2	26,872	0
BB	vs. Diuretic	1.629 (0.890–2.982)	0.113	2	10,883	68.3
CCB+Diuretic		1.578 (0.631–3.945)	0.329	1	414	NA
ACEi+CCB	vs. ACEi+Diuretic	0.842 (0.653–1.085)	0.184	1	11,506	NA
ARB+CCB	vs. CCB+Diuretic	1.402 (0.667–2.950)	0.373	1	2,204	NA
BB+CCB		2.292 (1.155–4.549)	0.018	1	2,183	NA
ARB+CCB	vs. ARB+Diuretic	0.955 (0.673–1.355)	0.798	1	5,141	NA
BB+CCB	vs. ARB+CCB	1.635 (0.886–3.016)	0.116	1	2,199	NA
Safety						
CCB	vs. Placebo	2.414 (2.006–2.906)	<0.001	1	7,665	NA
ARB		1.141 (1.081–1.204)	<0.001	4	39,022	89.9
ACEi		1.682 (1.271–2.226)	<0.001	7	31,306	94.4
Diuretic		1.981 (1.625–2.416)	<0.001	1	4,736	NA
ACEi+Diuretic		2.481 (2.007–3.067)	<0.001	1	11,140	NA
ARB+Diuretic		0.960 (0.886–1.041)	0.327	1	12,705	NA
Renin inhibitor		1.475 (1.250–1.741)	<0.001	1	8,561	NA
ACEi	vs. CCB	4.201 (2.206–7.998)	<0.001	1	1,748	NA
Diuretic		1.000 (0.516–1.938)	1.000	1	1,414	NA
CCB+Diuretic	vs. Diuretic	0.677 (0.236–1.937)	0.467	1	414	NA
ACEi+CCB	vs. ACEi+Diuretic	0.892 (0.823–0.966)	0.005	1	11,506	NA

An odds ratios <1 favor the former intervention and an odds ratios >1 favor the latter intervention.
CI, confidence interval; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; ACEi, angiotensin-converting enzyme inhibitor; BB, beta-blocker; NA, not applicable.

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Supplementary Table 3. Summary of results from pairwise meta-analysis and network meta-analysis for stroke prevention from randomized controlled trials

Comparison		No. of trials	OR (95% CI)	
			Pairwise meta-analysis	Network meta-analysis
Efficacy				
CCB	vs. Placebo	7	0.642 (0.539–0.765)	0.74 (0.67–0.82)
ARB		13	0.919 (0.864–0.978)	0.81 (0.73–0.89)
ACEi		13	0.813 (0.747–0.885)	0.81 (0.73–0.90)
Diuretic		8	0.632 (0.556–0.718)	0.68 (0.59–0.77)
BB		7	0.791 (0.673–0.929)	0.90 (0.78–1.03)
ACEi+THZ		1	0.986 (0.814–1.195)	0.99 (0.71–1.37)
CCB+THZ		1	0.698 (0.574–0.850)	0.71 (0.53–0.94)
ARB+THZ		1	0.795 (0.585–1.078)	0.78 (0.54–1.07)
Diuretic/BB		1	0.532 (0.335–0.847)	0.79 (0.65–0.94)
Renin inhibitor		1	1.209 (0.954–1.531)	1.26 (0.85–1.79)
ARB	vs. CCB	8	0.977 (0.869–1.099)	1.09 (0.97–1.22)
ACEi		8	1.123 (1.008–1.252)	1.09 (0.97–1.23)
Diuretic		4	1.063 (0.947–1.194)	0.91 (0.78–1.05)
BB		4	1.261 (1.116–1.425)	1.21 (1.05–1.39)
Diuretic/BB		3	0.922 (0.762–1.114)	1.06 (0.88–1.25)
ACEi	vs. ARB	5	1.074 (0.972–1.186)	1.00 (0.89–1.12)
BB		1	1.361 (1.142–1.622)	1.12 (0.95–1.29)
ARB+ACEi		3	1.017 (0.908–1.139)	0.97 (0.79–1.18)
ARB+CCB		1	0.606 (0.315–1.168)	0.89 (0.58–1.30)
Diuretic	vs. ACEi	1	0.871 (0.771–0.984)	0.84 (0.71–0.97)
BB		1	0.988 (0.546–1.789)	1.11 (0.95–1.30)
Diuretic/BB		2	1.097 (0.911–1.321)	0.98 (0.80–1.17)
ARB+ACEi		2	0.909 (0.813–1.017)	0.97 (0.78–1.19)
BB	vs. diuretic	2	1.629 (0.890–2.982)	1.34 (1.11–1.58)
CCB+THZ		1	1.578 (0.631–3.945)	1.06 (0.76–1.44)
ACEi+CCB	vs. ACEi+THZ	1	0.842 (0.653–1.085)	0.85 (0.58–1.21)
ARB+CCB	vs. CCB+THZ	1	1.402 (0.667–2.950)	1.02 (0.64–1.56)
BB+CCB		1	2.292 (1.155–4.549)	2.00 (1.03–3.61)
ARB+CCB	vs. ARB+THZ	1	0.955 (0.673–1.355)	0.93 (0.64–1.34)
BB+CCB	vs. ARB+CCB	1	1.635 (0.886–3.016)	1.96 (1.04–3.58)
Tolerability				
CCB	vs. Placebo	1	2.414 (2.006–2.906)	1.43 (0.53–3.09)
ARB		4	1.141 (1.081–1.204)	1.10 (0.54–2.01)
ACEi		7	1.682 (1.271–2.226)	2.15 (1.30–3.52)
Diuretic		1	1.981 (1.625–2.416)	1.86 (0.56–4.59)
ACEi+THZ		1	2.481 (2.007–3.067)	3.10 (0.64–9.27)
ARB+THZ		1	0.960 (0.886–1.041)	1.20 (0.26–3.51)
Renin inhibitor		1	1.475 (1.250–1.741)	1.89 (0.40–5.58)
ACEi	vs. CCB	1	4.201 (2.206–7.998)	1.45 (0.42–3.71)
Diuretic		1	1.000 (0.516–1.938)	1.45 (0.42–3.71)
CCB+THZ	vs. diuretic	1	0.677 (0.236–1.937)	0.95 (0.12–3.46)
ACEi+CCB	vs. ACEi+THZ	1	0.892 (0.823–0.966)	1.14 (0.24–3.38)

OR, odds ratio; CI, confidence interval; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; ACEi, angiotensin-converting enzyme inhibitor; BB, beta-blocker; THZ, thiazide-like diuretic.

Supplementary Table 4. Results of sensitivity analyses for efficacy

Drug name	Standard analysis		Published in 2000 or later only		Duration of follow-up longer than 3 years		Trials including hypertensive participants only		Trials excluding hearts failure participants		Age ≤60 years		Age >60 years		Participants without history of stroke		Participants with history of stroke		Participants without history of DM		Participants with history of DM		Participants with baseline SBP ≤150 mm Hg		Participants with baseline SBP >150 mm Hg	
	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank	OR (95% CI)	SUCRA rank
CCB	0.74 (0.67–0.82)	4	0.83 (0.72–0.94)	6	0.75 (0.66–0.86)	4	0.64 (0.54–0.74)	5	0.73 (0.65–0.82)	4	0.80 (0.56–1.11)	4	0.74 (0.66–0.82)	3	0.86 (0.66–1.09)	5	-	-	0.74 (0.30–1.35)	7	-	-	0.76 (0.63–0.92)	1	0.64 (0.54–0.75)	5
ARB	0.81 (0.73–0.89)	9	0.84 (0.76–0.9)	8	0.80 (0.71–0.90)	8	0.61 (0.49–0.73)	3	0.80 (0.71–0.89)	9	0.94 (0.60–1.39)	6	0.83 (0.75–0.91)	9	0.84 (0.66–1.02)	3	-	-	0.62 (0.23–1.28)	4	-	-	0.88 (0.79–0.98)	8	0.65 (0.53–0.78)	6
ACEi	0.81 (0.73–0.90)	10	0.85 (0.76–0.94)	9	0.80 (0.70–0.90)	7	0.69 (0.55–0.85)	8	0.78 (0.69–0.89)	8	0.75 (0.52–0.99)	2	0.83 (0.75–0.92)	10	0.86 (0.68–1.02)	4	-	-	0.54 (0.20–1.05)	2	-	-	0.86 (0.76–0.95)	7	0.64 (0.48–0.84)	3
Diuretic	0.68 (0.59–0.77)	1	0.79 (0.66–0.94)	3	0.65 (0.55–0.77)	1	0.60 (0.49–0.72)	1	0.66 (0.56–0.76)	1	0.39 (0.22–0.60)	1	0.72 (0.63–0.82)	1	0.62 (0.40–0.86)	1	-	-	0.41 (0.17–0.72)	1	-	-	0.78 (0.60–0.99)	2	0.63 (0.53–0.74)	2
BB	0.90 (0.78–1.03)	11	1.03 (0.83–1.24)	12	0.90 (0.76–1.06)	11	0.79 (0.66–0.93)	9	0.88 (0.75–1.02)	11	0.90 (0.67–1.21)	5	0.94 (0.78–1.11)	12	1.01 (0.59–1.62)	8	-	-	0.66 (0.36–0.98)	6	-	-	0.84 (0.61–1.12)	6	0.83 (0.69–0.98)	9
ACEi+ diuretic	0.99 (0.71–1.37)	12	1.00 (0.71–1.36)	11	1.00 (0.70–1.40)	12	1.08 (0.79–1.41)	13	1.01 (0.68–1.43)	12	-	-	1.01 (0.74–1.33)	13	-	-	-	-	-	-	-	-	0.99 (0.73–1.32)	9	-	-
CCB+ diuretic	0.71 (0.53–0.94)	2	0.69 (0.50–0.91)	1	0.70 (0.51–0.95)	2	0.68 (0.47–0.91)	6	0.70 (0.50–0.96)	2	-	-	0.71 (0.54–0.92)	2	1.15 (0.35–2.60)	7	-	-	0.87 (0.12–2.79)	5	-	-	-	-	0.68 (0.48–0.92)	7
ARB+ diuretic	0.78 (0.54–1.07)	5	0.78 (0.55–1.08)	4	0.77 (0.53–1.10)	5	0.559 (0.27–1.15)	4	0.78 (0.53–1.12)	7	-	-	0.78 (0.54–1.07)	5	0.83 (0.53–1.20)	2	-	-	-	-	-	-	0.80 (0.54–1.15)	3	0.65 (0.30–1.22)	4
Diuretic /BB	0.79 (0.65–0.94)	7	0.87 (0.65–1.13)	10	0.82 (0.65–1.02)	10	0.68 (0.53–0.85)	7	0.77 (0.62–0.94)	5	0.77 (0.30–1.60)	3	0.79 (0.65–0.94)	6	0.93 (0.68–1.19)	6	-	-	0.64 (0.10–1.88)	3	-	-	-	-	0.67 (0.52–0.84)	8
ARB+ ACEi	0.78 (0.63–0.96)	6	0.81 (0.65–0.98)	5	0.78 (0.57–1.04)	6	-	-	0.77 (0.57–1.02)	6	-	-	0.80 (0.66–0.96)	7	-	-	-	-	-	-	-	-	0.83 (0.68–1.00)	4	-	-
ARB+ CCB	0.72 (0.47–1.05)	3	0.72 (0.48–1.05)	2	0.71 (0.47–1.03)	3	0.56 (0.31–0.94)	2	0.71 (0.46–1.06)	3	-	-	0.72 (0.49–1.05)	4	1.81 (0.36–4.88)	10	-	-	-	-	-	-	-	-	0.60 (0.33–0.99)	1
ACEi+ CCB	0.84 (0.50–1.34)	8	0.86 (0.51–1.36)	7	0.86 (0.50–1.39)	9	1.16 (0.68–1.34)	12	0.86 (0.47–1.44)	10	-	-	0.86 (0.54–1.30)	8	-	-	-	-	-	-	-	-	0.85 (0.52–1.29)	5		-
BB+ CCB	1.38 (0.72–2.53)	14	1.40 (0.71–2.49)	14	1.42 (0.68–2.61)	14	0.21 (0.58–2.25)	10	1.40 (0.68–2.58)	14	-	-	1.42 (0.76–2.44)	15	2.76 (0.63–7.84)	11	-	-	-	-	-	-	-	-	1.27 (0.61–2.32)	11
Renin inhibitor	1.26 (0.85–1.79)	15	1.23 (0.85–1.73)	15	-	-	-	-	-	-	-	-	0.82 (0.35–1.64)	11	-	-	-	-	-	-	-	-	1.23 (0.88–1.68)	11		-

The table shows the effect sizes (ORs) and the rank order (SUCRA ranks) compared to placebo before (standard analysis) and after sensitivity analyses. In subgroup of participants without history of stroke, most of the results were stable except for ARB+CCB. Sensitivity analyses were not feasible for two subgroups (participants with history of stroke and participants with history of DM) due to limited number of trials. Sensitivity and subgroup analyses on tolerability were not feasible due to the low event rate and few trials reported such endpoint.

DM, diabetes mellitus; SBP, systolic blood pressure; OR, odds ratio; CI, confidence interval; SUCRA, surface under the cumulative ranking curve; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; ACEi, angiotensin-converting enzyme inhibitor; BB, beta-blocker.

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Supplementary Table 5. Inconsistency test by node-splitting for efficacy

Side	Direct		Indirect		Difference		*P*>|z|	tau
	Coefficient	SE	Coefficient	SE	Coefficient	SE		
A B	−0.451284	0.1068399	−0.2510002	0.0631098	−0.2002838	0.1243141	0.107	0.1242378
A C	−0.0993669	0.0520806	−0.3976046	0.0644907	0.2982377	0.0829809	0.000	0.0888897
A D	−0.1920362	0.0714549	−0.232732	0.076429	0.0406958	0.1053677	0.699	0.1283594
A E	−0.4643451	0.0847979	−0.2713766	0.1006676	−0.1929685	0.1293544	0.136	0.1182247
A F	−0.2418242	0.0961056	0.0165464	0.0910915	−0.2583706	0.1313758	0.050	0.1158177
A G*	−0.0140438	0.1571743	0.0781284	217.4085	−0.0921723	217.4086	1.000	0.1228489
A H	−0.3589984	0.1597569	−0.3847499	0.334987	0.0257515	0.3711314	0.945	0.1243285
A I	−0.2299547	0.1994513	−0.3698523	0.3509167	0.1398976	0.4036376	0.729	0.1244593
A J	−0.6301822	0.2651127	−0.1922366	0.0978299	−0.4379456	0.2825871	0.121	0.119769
A 0	-	-	-	-	-	-	-	-
B C	−0.0701689	0.0855201	0.1973536	0.0743317	−0.2675225	0.1141232	0.019	0.120526
B D	0.0495561	0.0963298	0.1198465	0.0790357	−0.0702904	0.1255804	0.576	0.1287696
B E	0.0605161	0.1030118	−0.2094289	0.0979282	0.269945	0.1424187	0.058	0.1165115
B F	0.2197186	0.105716	0.1710801	0.0985393	0.0486385	0.1449476	0.737	0.127049
B J	0.0822924	0.0974244	−0.0422592	0.1880368	0.1245516	0.2117605	0.556	0.1264311
C D	0.0514294	0.0883482	−0.0205073	0.0719231	0.0719367	0.1138033	0.527	0.124331
C F	0.3083339	0.1486445	0.0536351	0.0817897	0.2546988	0.1696607	0.133	0.1186854
C K*	0.0172791	0.1024207	−0.3017673	0.2237688	0.3190464	0.246183	0.195	0.1211038
C L	−0.5002282	0.3563764	0.0453293	0.2466586	−0.5455575	0.4334104	0.208	0.1229408
D E	−0.1350791	0.1418873	−0.1944038	0.0911882	0.0593247	0.1688091	0.725	0.1276075
D F	−0.012024	0.3272441	0.1095738	0.0798881	−0.1215978	0.3368545	0.718	0.1237679
D J	0.081128	0.1434105	−0.1187521	0.1217839	0.1998801	0.1881917	0.288	0.1223478
D K	−0.1067001	0.105547	0.1686203	0.1995086	−0.2753204	0.2250074	0.221	0.1207032
E F	0.3827491	0.1882641	0.2493307	0.1005048	0.1334183	0.2139697	0.533	0.1259243
E H	0.4562378	0.483335	−0.0271547	0.1642594	0.4833925	0.510484	0.344	0.1229657
G M*	−0.1723831	0.1785902	0.0668541	438.5716	−0.2392372	438.5717	1.000	0.122849
H L*	0.3381917	0.399266	−0.139543	0.2746922	0.4777347	0.484633	0.324	0.1241665
H N*	0.8295874	0.3710168	−0.125879	0.8421779	0.9554665	0.9692651	0.324	0.1241662
I L	−0.0457219	0.2175149	−0.1856204	0.3400159	0.1398984	0.4036379	0.729	0.1244593
L N*	0.4913958	0.3363546	1.446865	0.8847719	−0.9554689	0.9692654	0.324	0.1241666

SE, standard error; A, placebo; B, calcium channel blocker (CCB); C, angiotensin II receptor blocker (ARB); D, angiotensin-converting enzyme inhibitor (ACEi); E, diuretic; F, beta-blocker (BB); G, ACEi+diuretic; H, CCB+diuretic; I, ARB+diuretic; J, BB/diuretic; K, ARB+ACEi; L, ARB+CCB; M, ACEi+CCB; N, BB+CCB.
*Warning: all the evidence about these contrasts comes from the trials which directly compare them.

Supplementary Table 6. Inconsistency test by node-splitting for tolerability

Side	Direct		Indirect		Difference		*P*>|z|	tau
	Coefficient	SE	Coefficient	SE	Coefficient	SE		
A B	0.8814234	0.4786115	−0.2520848	0.47102	1.133508	0.6715123	0.091	0.4691702
A C	-	-	-	-	-	-	-	-
A D	0.5349264	0.1638938	2.2638	0.5903911	−1.728874	0.613268	0.005	0.3675151
A E	0.6838528	0. 5614273	0.190057	0.7790083	0.4937958	0.9602367	0.607	0.5522498
A F	-	-	-	-	-	-	-	-
A G*	0.9085095	0. 5370918	0.0566346	69.0073	0.8518749	69.00939	0.990	0.5260806
A I	-	-	-	-	-	-	-	-
A K	-	-	-	-	-	-	-	-
B D	1.435238	0.4929838	−0.2936447	0.3647816	1.728883	0.6132688	0.005	0.367515
B E	7.31E-10	0.6472978	0.4937874	0.7092778	−0.4937874	0.9602445	0.607	0.5522505
E H*	−0.3904272	0.7513723	−1.078526	338.4009	0.6880987	338.4016	0.998	0.5260769
G J*	−0.1147697	0.5276537	−1.818516	138.0747	1.703746	138.0757	0.990	0.5260807

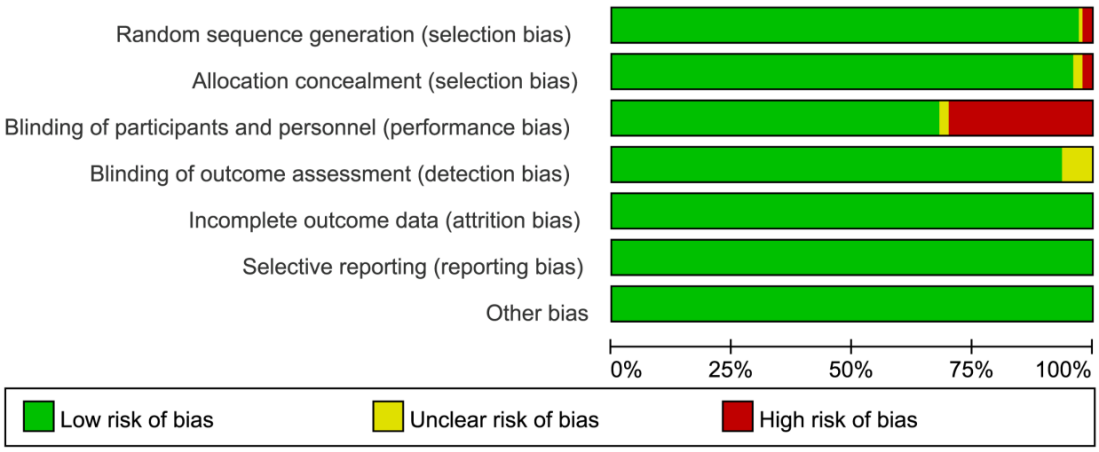
SE, standard error; A, placebo; B, calcium channel blocker (CCB); C, angiotensin II receptor blocker (ARB); D, angiotensin-converting enzyme inhibitor (ACEi); E, diuretic; F, beta-blocker (BB); G, ACEi+diuretic; H, CCB+diuretic; I, ARB+diuretic; J, BB/diuretic; K, ARB+ACEi.
*Warning: all the evidence about these contrasts comes from the trials which directly compare them.

Supplementary Table 7. Inconsistency test by design-by-treatment for efficacy and tolerability

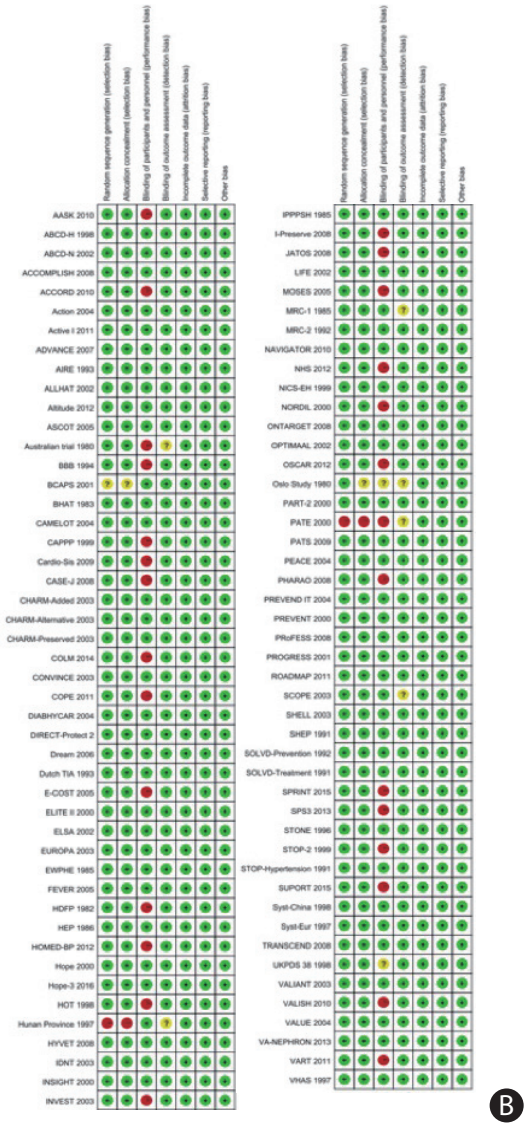
Network outcome	Chi-square	*P* for test of global inconsistency
Efficacy	31.35	0.26
Safety	7.14	0.03

Assessment of global inconsistency in networks for efficacy and tolerability in preventing stroke using the 'design-by-treatment' interaction model.

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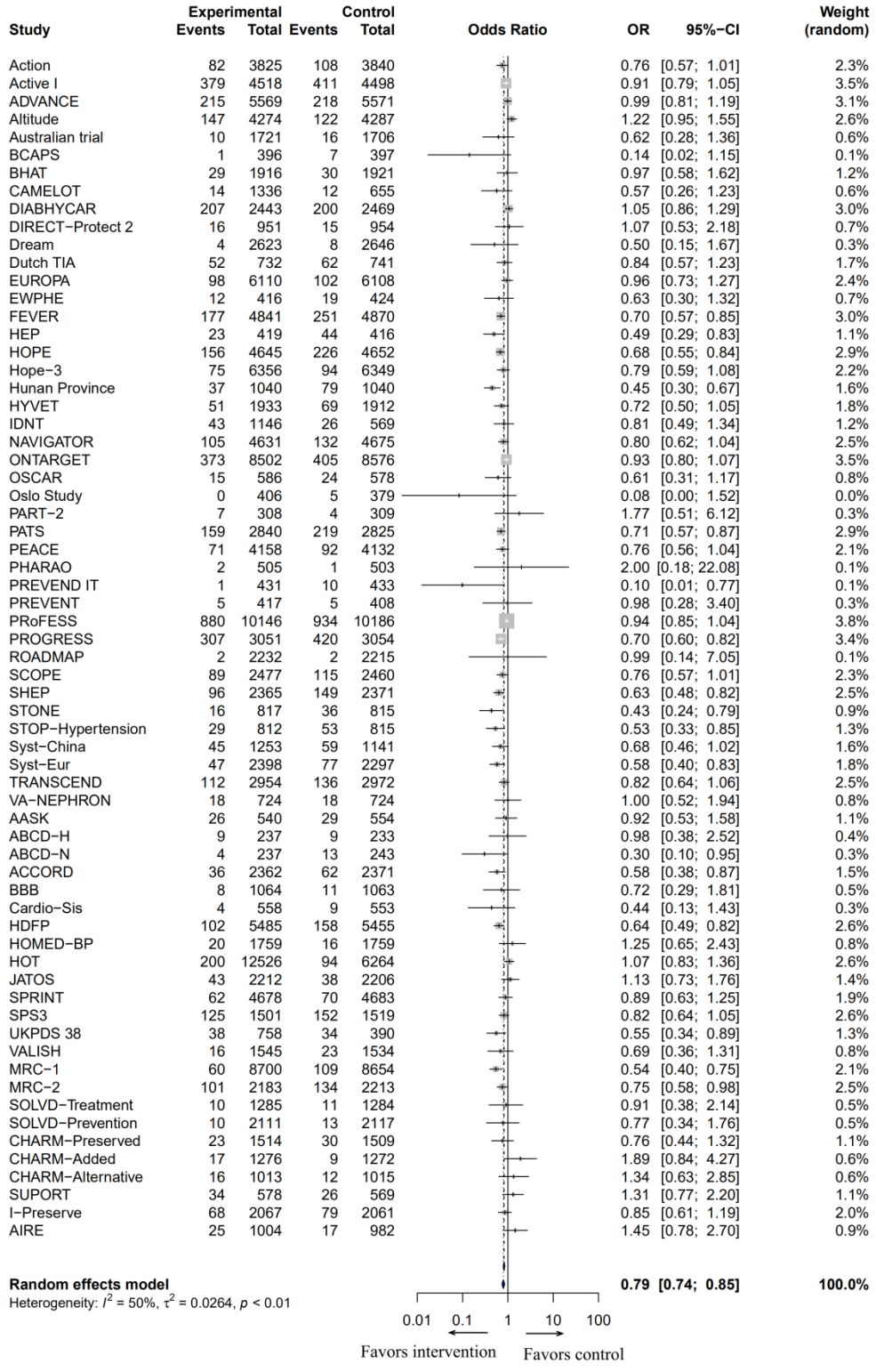


A



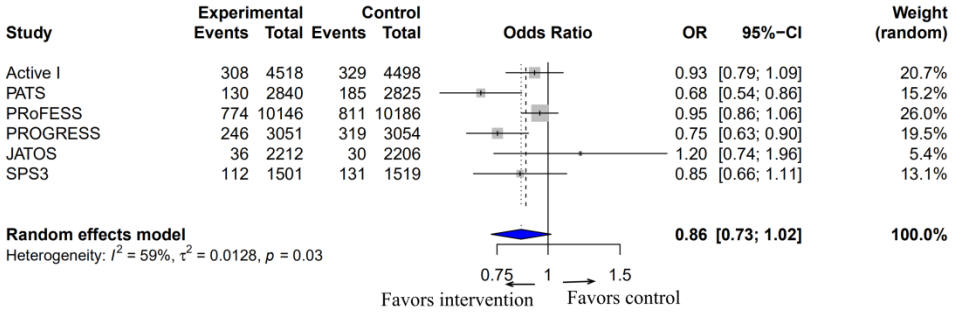
B

Supplementary Figure 1. Risk of bias of the included studies. (A) Risk of bias graph. (B) Risk of bias summary. "+" indicates low risk of bias; "?" indicates unclear risk of bias; "–" indicates high risk of bias.

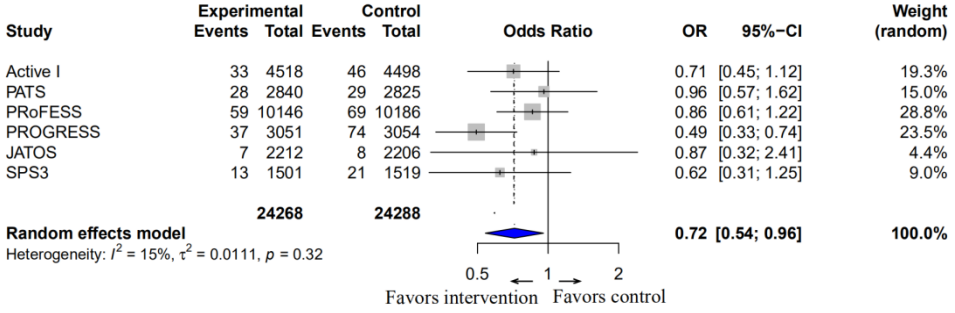


Supplementary Figure 2. Reduction in systolic blood pressure on the odds ratio (OR) of stroke. CI, confidence interval.

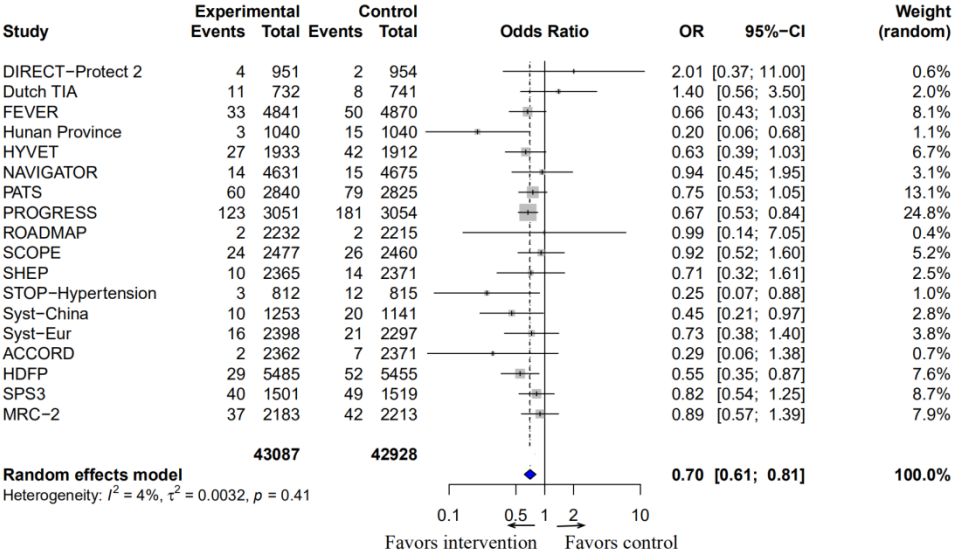
https://doi.org/10.5853/jos.2020.02698



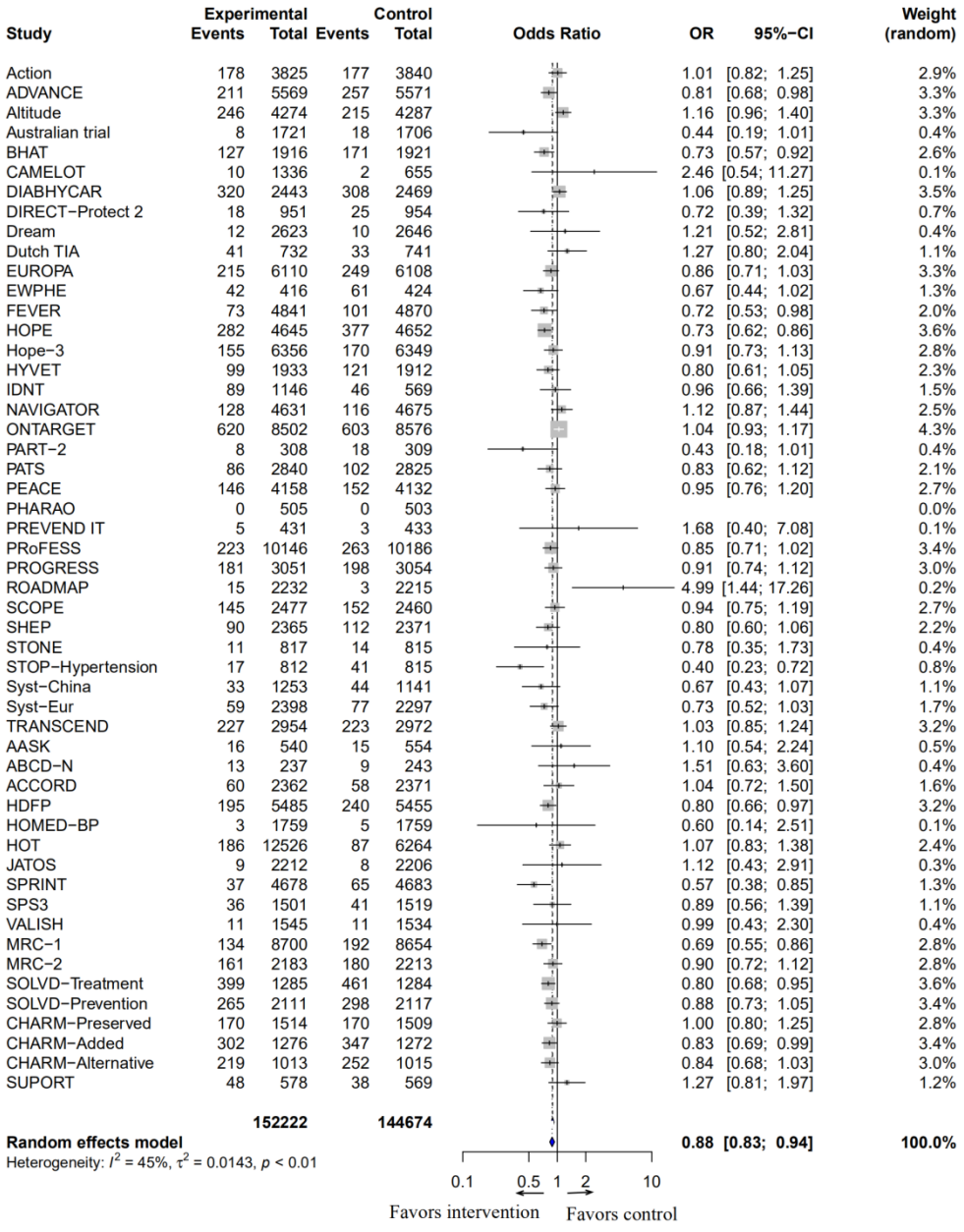
Supplementary Figure 3. Reduction in systolic blood pressure on the odds ratio (OR) of ischemic stroke. CI, confidence interval.



Supplementary Figure 4. Reduction in systolic blood pressure on the odds ratio (OR) of hemorrhagic stroke. CI, confidence interval.

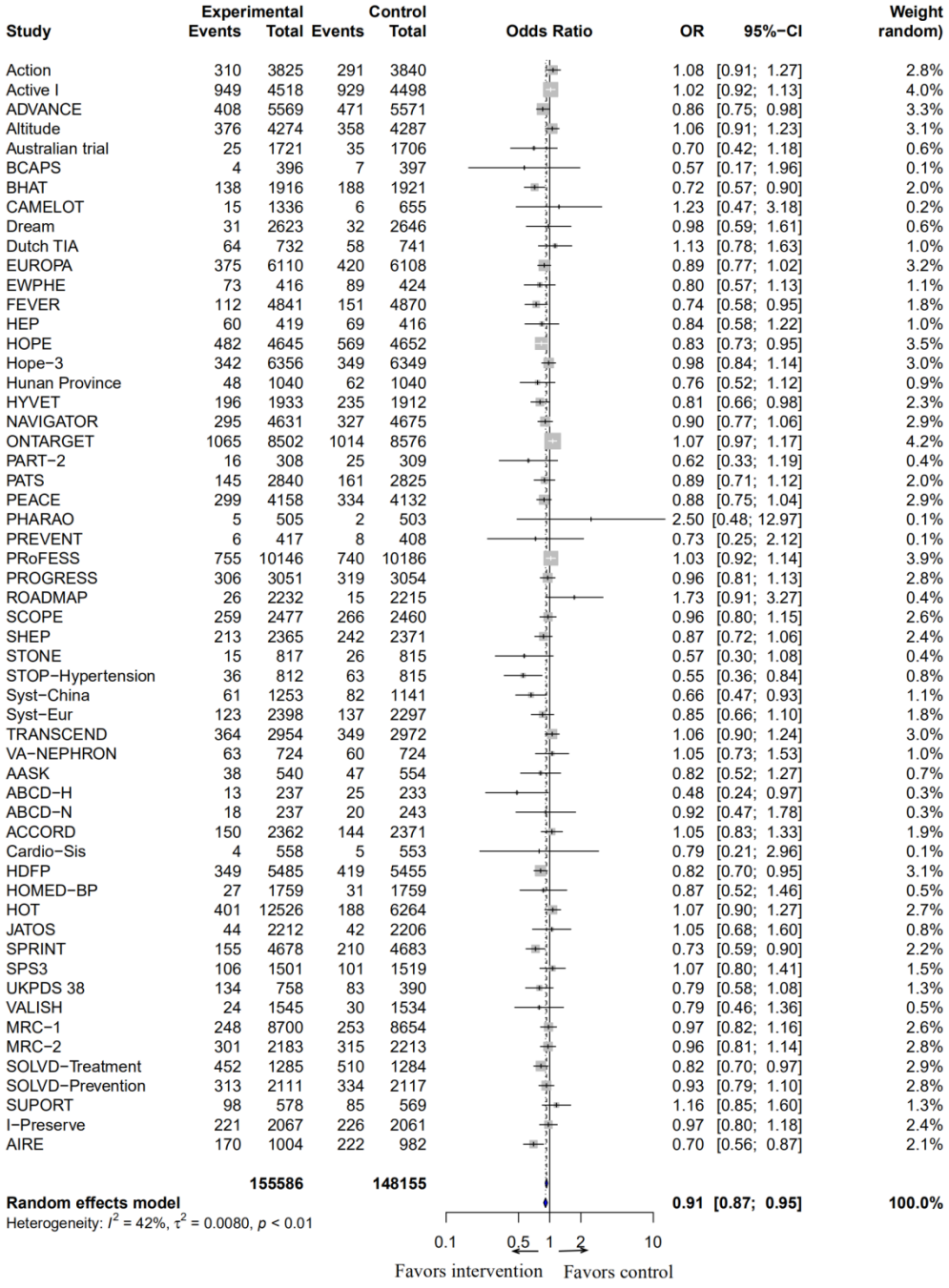


Supplementary Figure 5. Reduction in systolic blood pressure on the odds ratio (OR) of fatal or disabling stroke. CI, confidence interval.

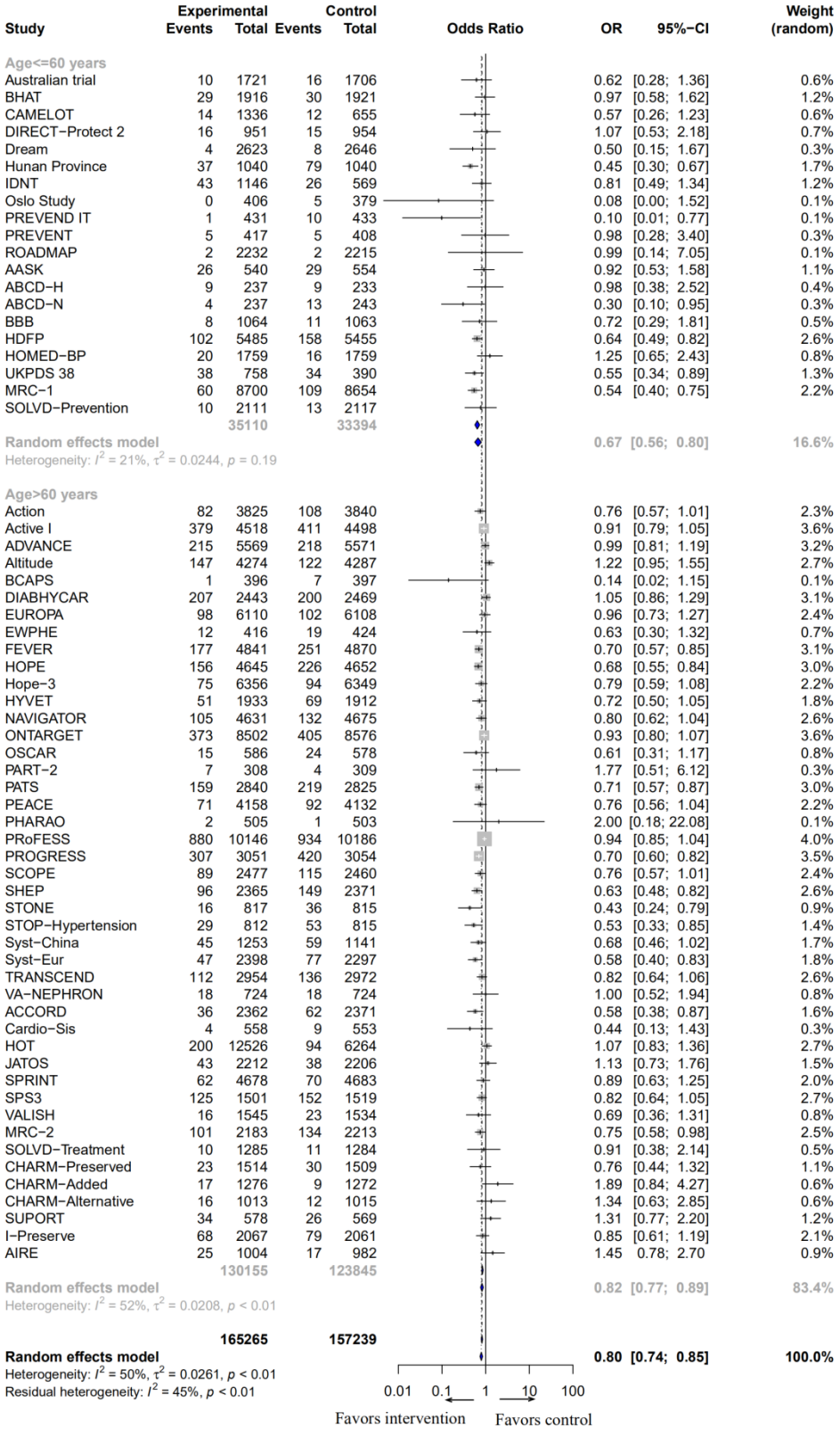


Supplementary Figure 6. Reduction in systolic blood pressure on the odds ratio (OR) of cardiovascular death. CI, confidence interval.

https://doi.org/10.5853/jos.2020.02698

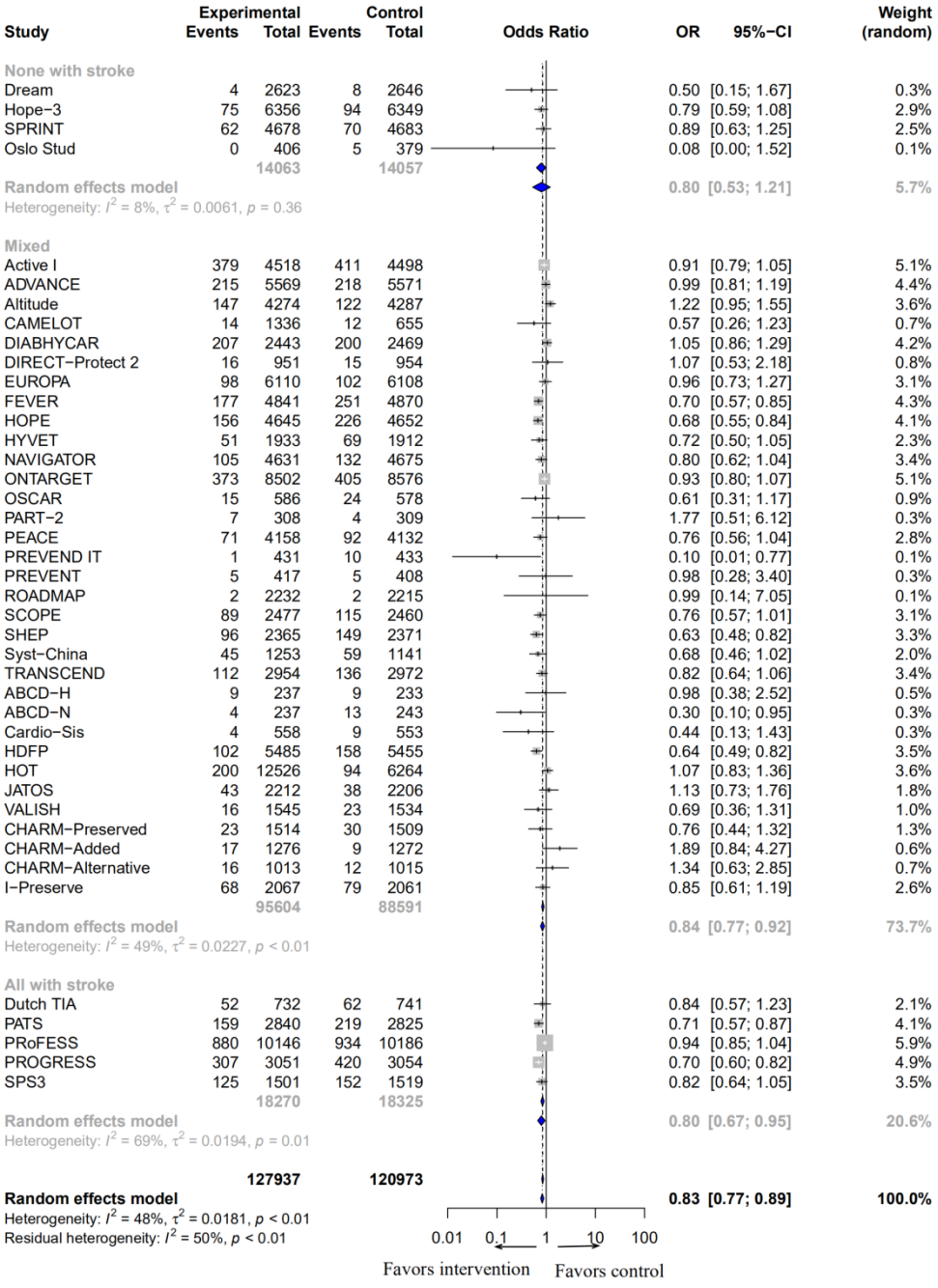


Supplementary Figure 7. Reduction in systolic blood pressure on the odds ratio (OR) of all cause death. CI, confidence interval.

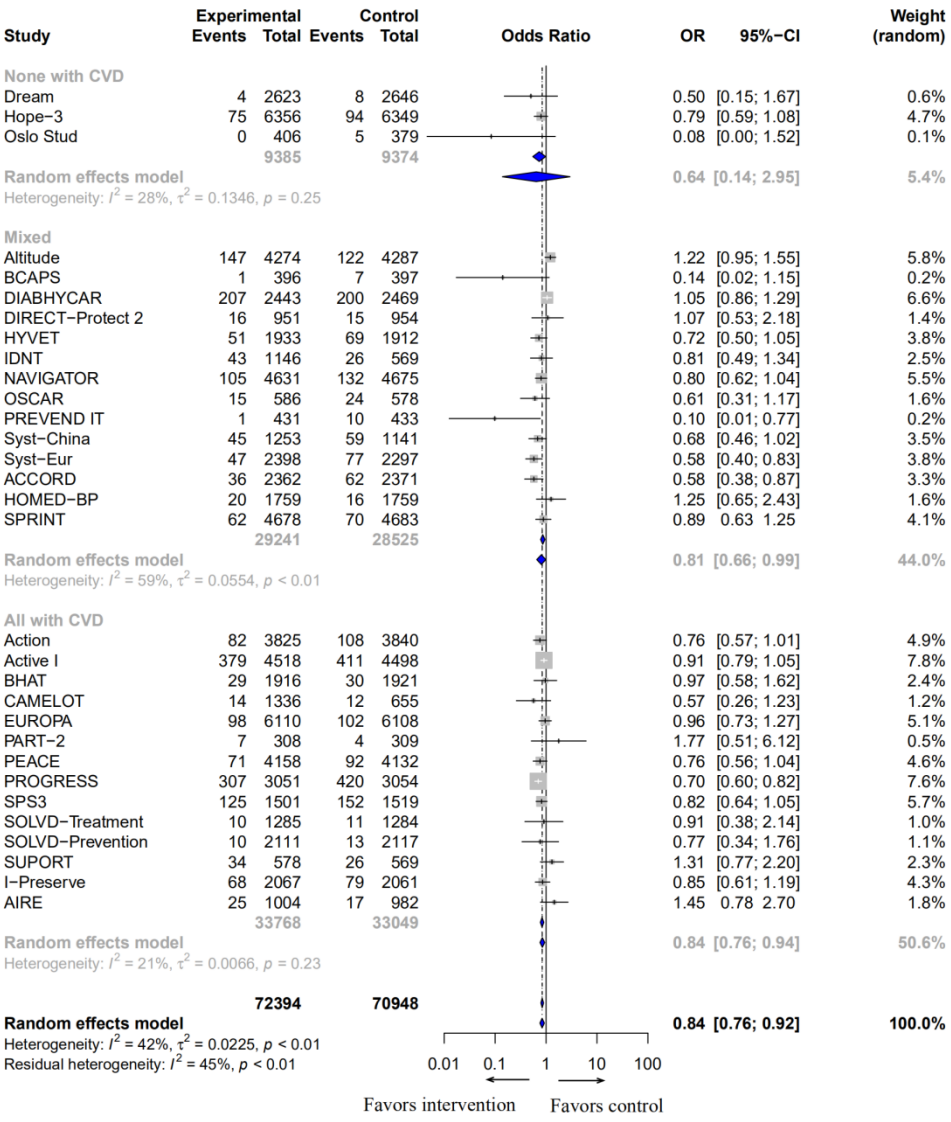


Supplementary Figure 8. Reduction in systolic blood pressure on the odds ratio (OR) of stroke stratified by age. CI, confidence interval.

https://doi.org/10.5853/jos.2020.02698

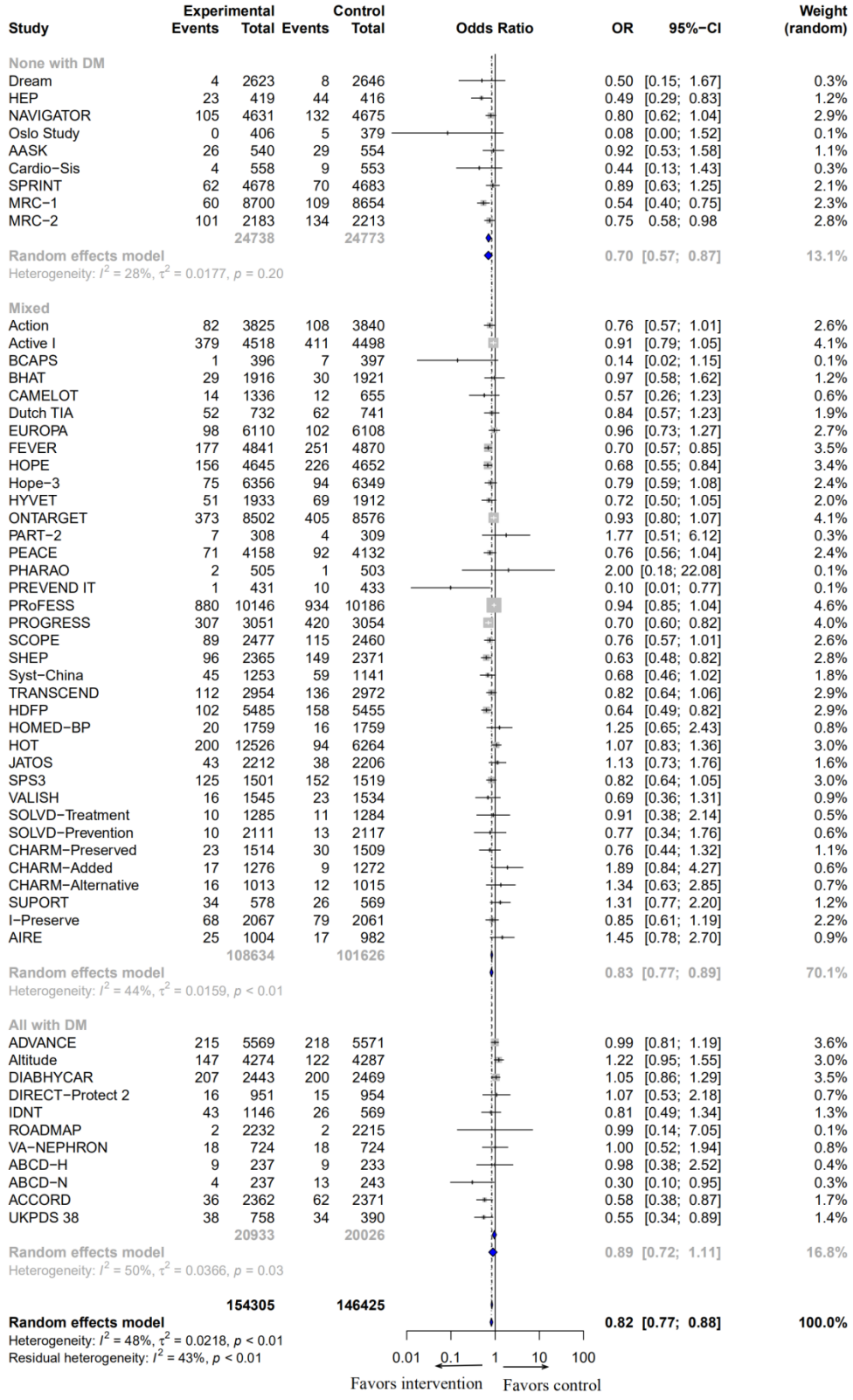


Supplementary Figure 9. Reduction in systolic blood pressure on the odds ratio (OR) of stroke stratified by history of stroke. CI, confidence interval.

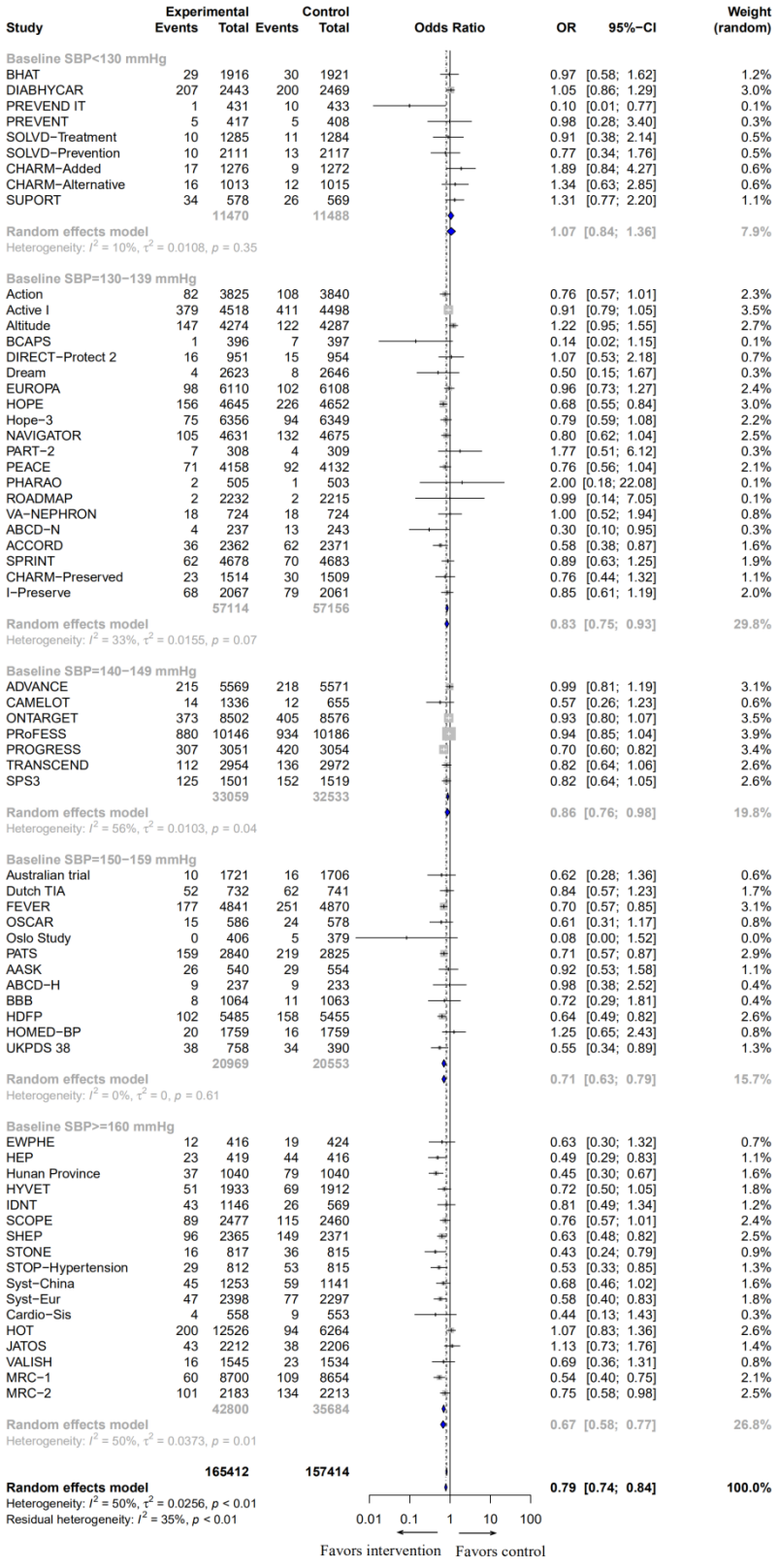


Supplementary Figure 10. Reduction in systolic blood pressure on the odds ratio (OR) of stroke stratified by history of cardiovascular disease (CVD). CI, confidence interval.

https://doi.org/10.5853/jos.2020.02698

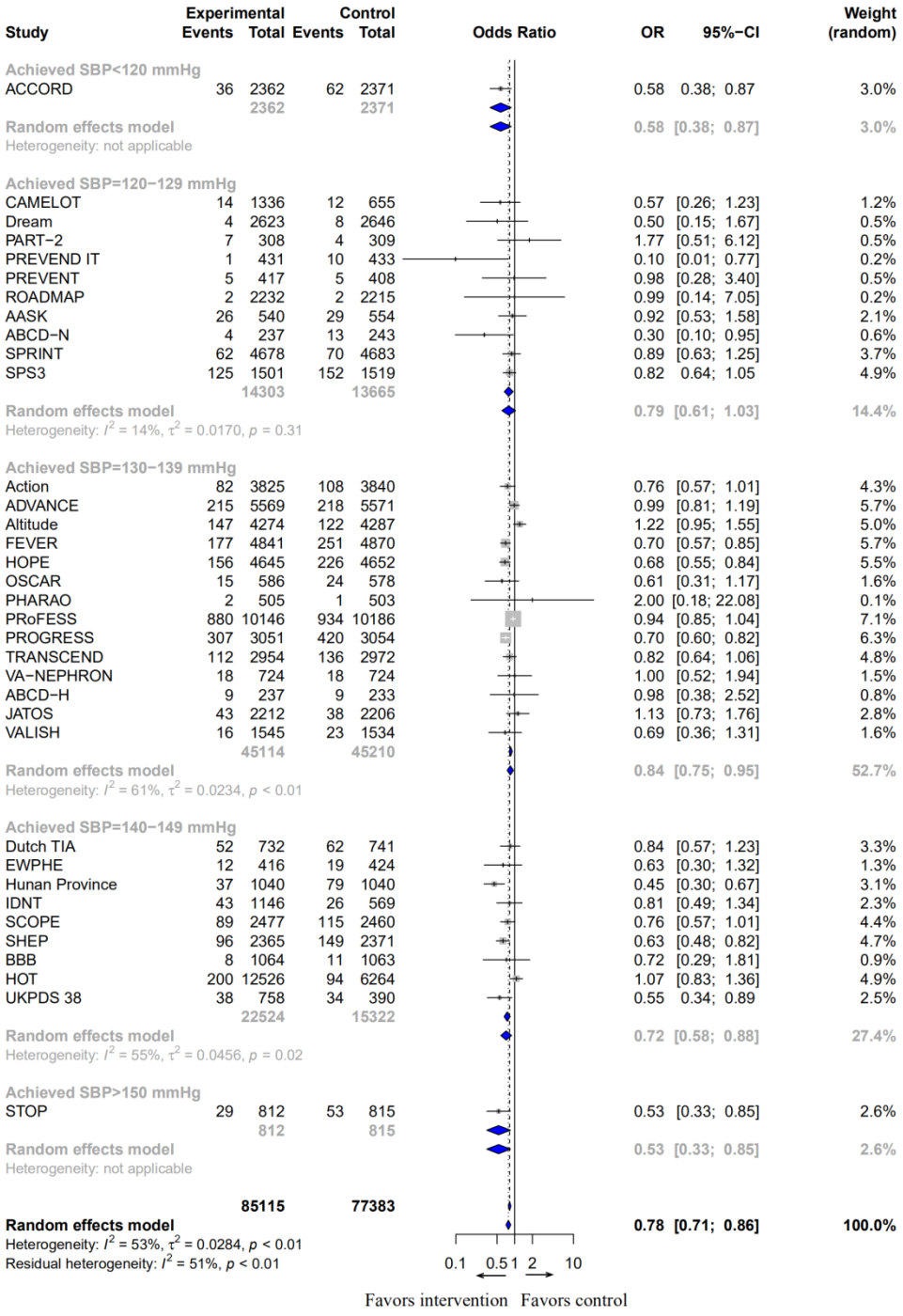


Supplementary Figure 11. Reduction in systolic blood pressure on the odds ratio (OR) of stroke stratified by history of diabetes mellitus (DM). CI, confidence interval.

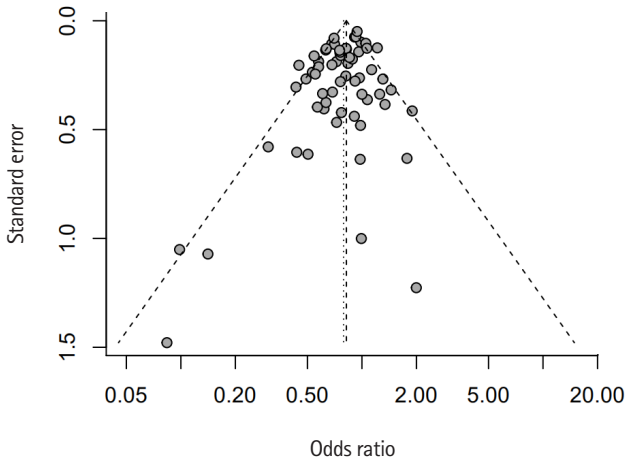


Supplementary Figure 12. Reduction in systolic blood pressure (SBP) on the odds ratio (OR) of stroke stratified by baseline SBP levels. CI, confidence interval.

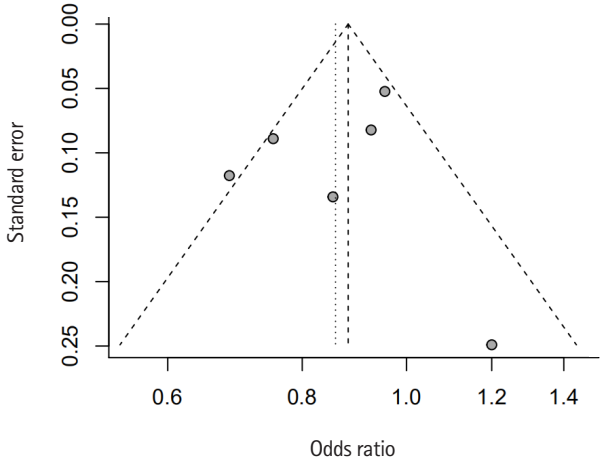
https://doi.org/10.5853/jos.2020.02698



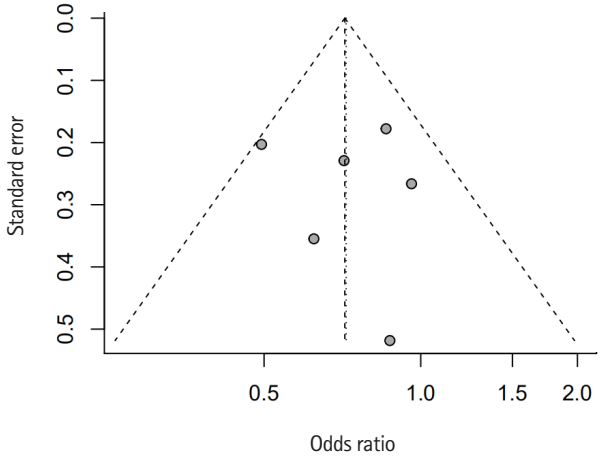
Supplementary Figure 13. Reduction in systolic blood pressure (SBP) on the odds ratio (OR) of stroke stratified by achieved SBP levels. CI, confidence interval.



Supplementary Figure 14. Funnel plot for stroke.

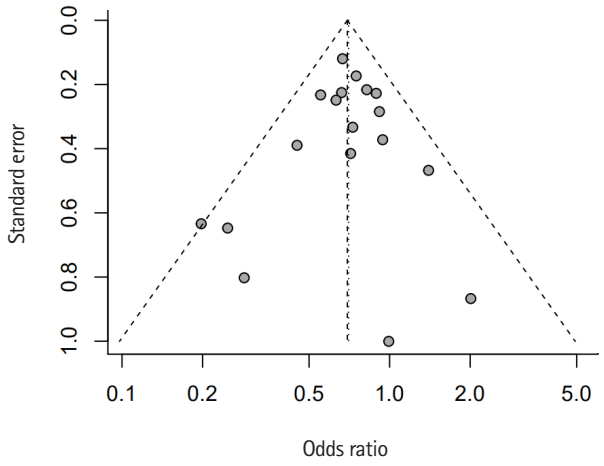


Supplementary Figure 15. Funnel plot for ischemic stroke.

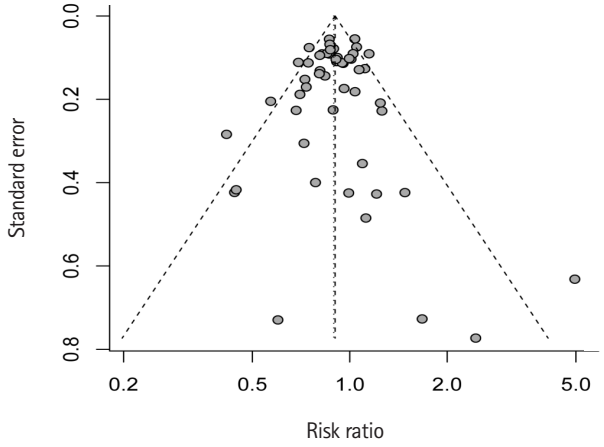


Supplementary Figure 16. Funnel plot for hemorrhagic stroke.

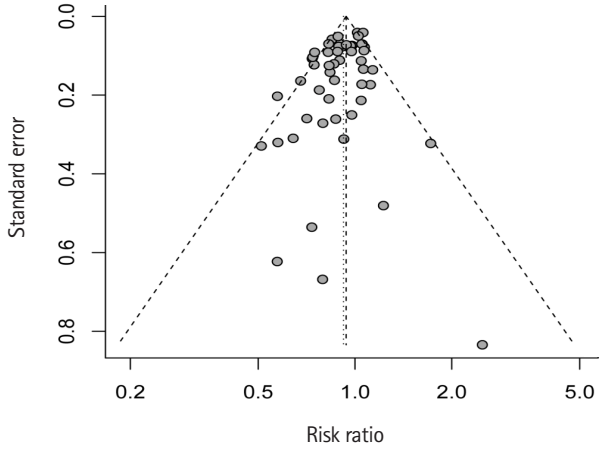
https://doi.org/10.5853/jos.2020.02698



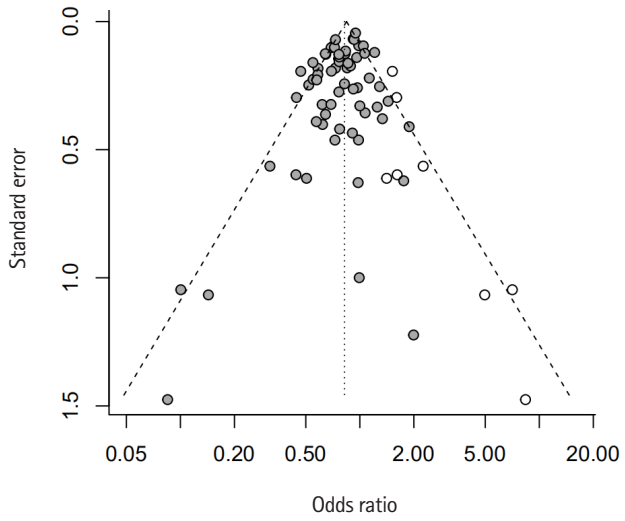
Supplementary Figure 17. Funnel plot for fatal or disabling stroke.



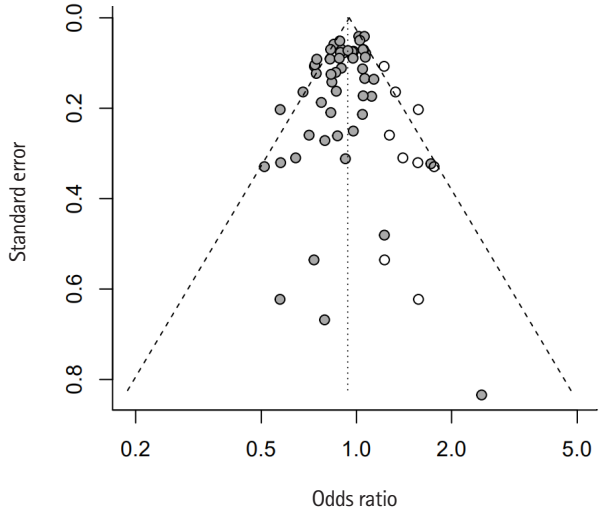
Supplementary Figure 18. Funnel plot for cardiovascular death.



Supplementary Figure 19. Funnel plot for all cause death.

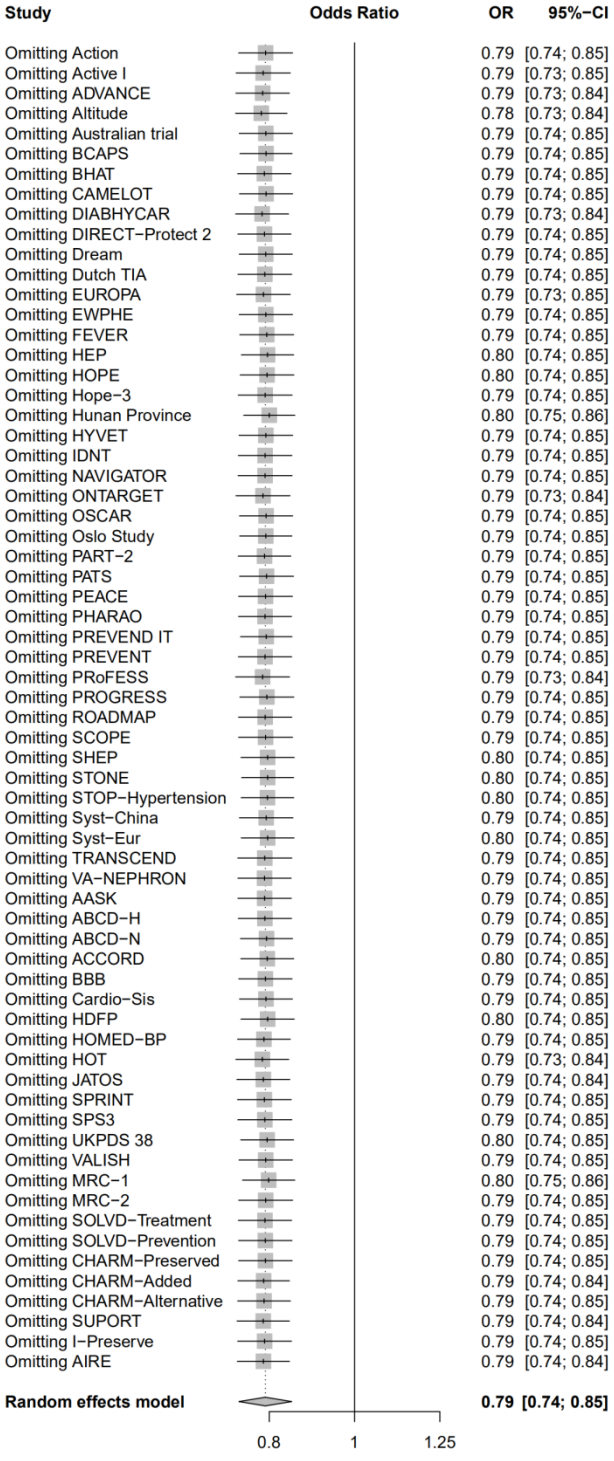


Supplementary Figure 20. Trimmed funnel plot for stroke.

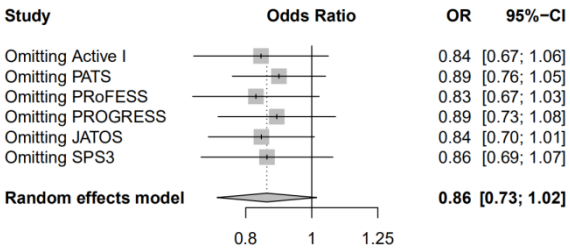


Supplementary Figure 21. Trimmed funnel plot for all cause death.

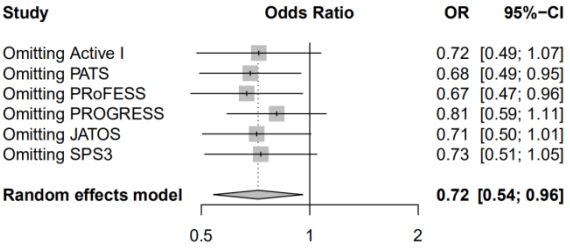
https://doi.org/10.5853/jos.2020.02698



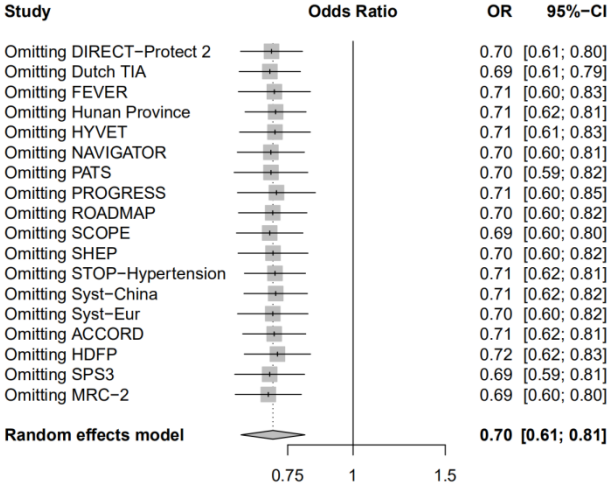
Supplementary Figure 22. Sensitivity analysis for stroke. OR, odds ratio; CI, confidence interval.



Supplementary Figure 23. Sensitivity analysis for ischemic stroke. OR, odds ratio; CI, confidence interval.

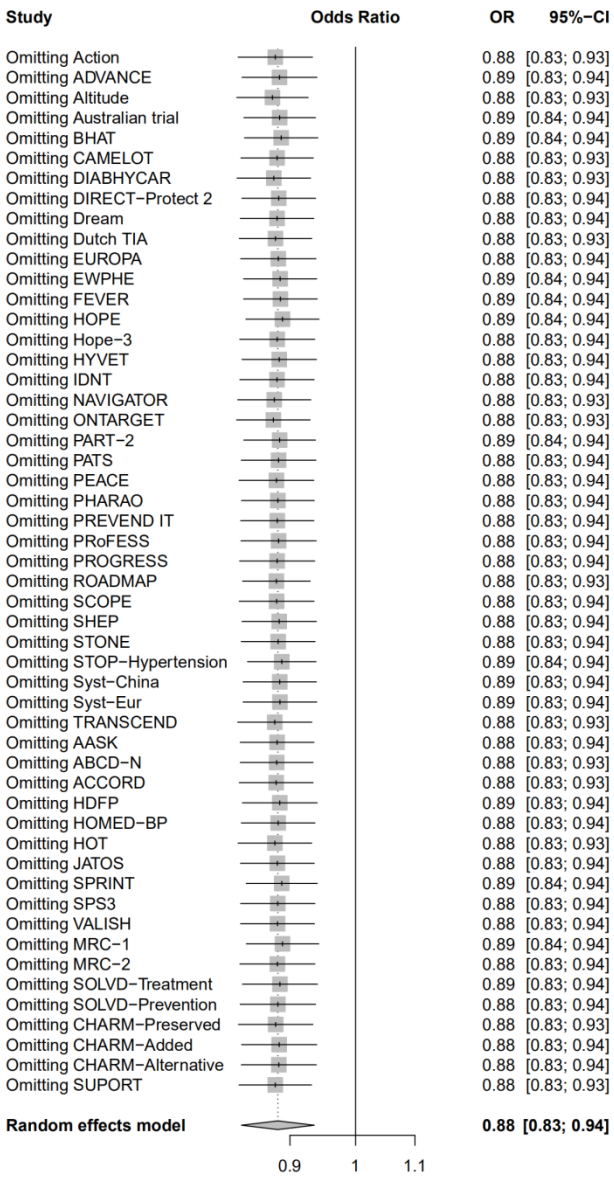


Supplementary Figure 24. Sensitivity analysis for hemorrhagic stroke. OR, odds ratio; CI, confidence interval.

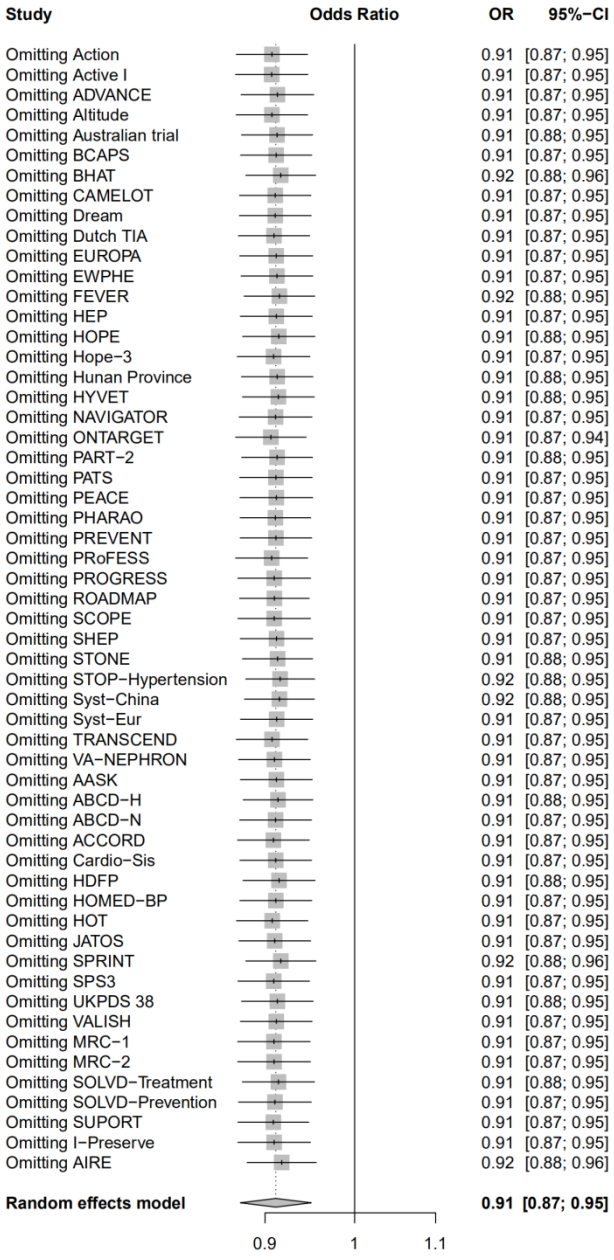


Supplementary Figure 25. Sensitivity analysis for fatal or disabling stroke. OR, odds ratio; CI, confidence interval.

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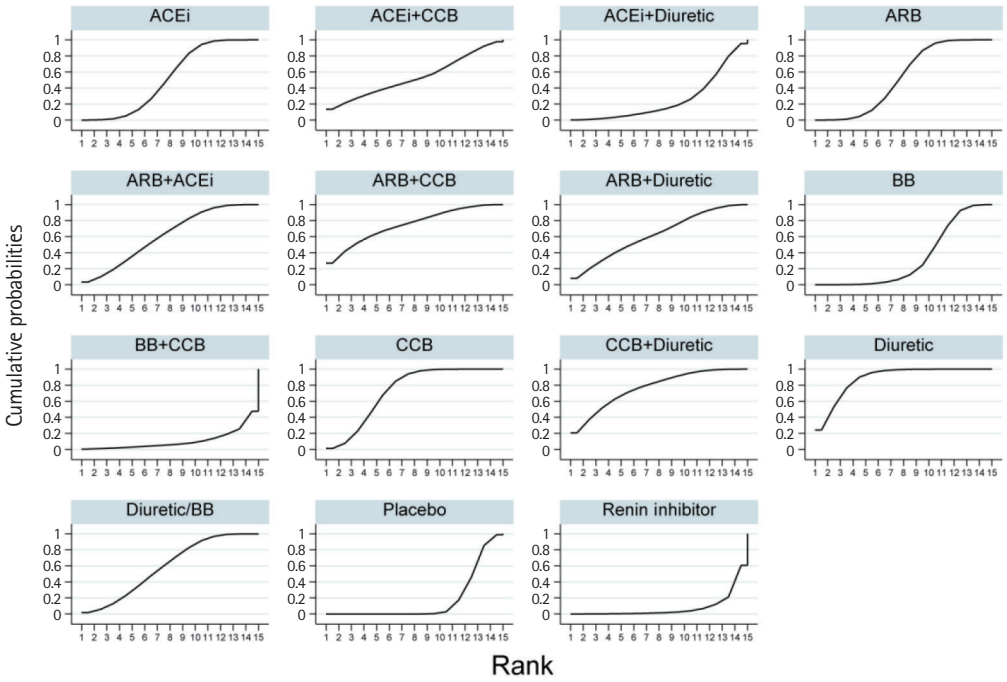


Supplementary Figure 26. Sensitivity analysis for cardiovascular death. OR, odds ratio; CI, confidence interval.

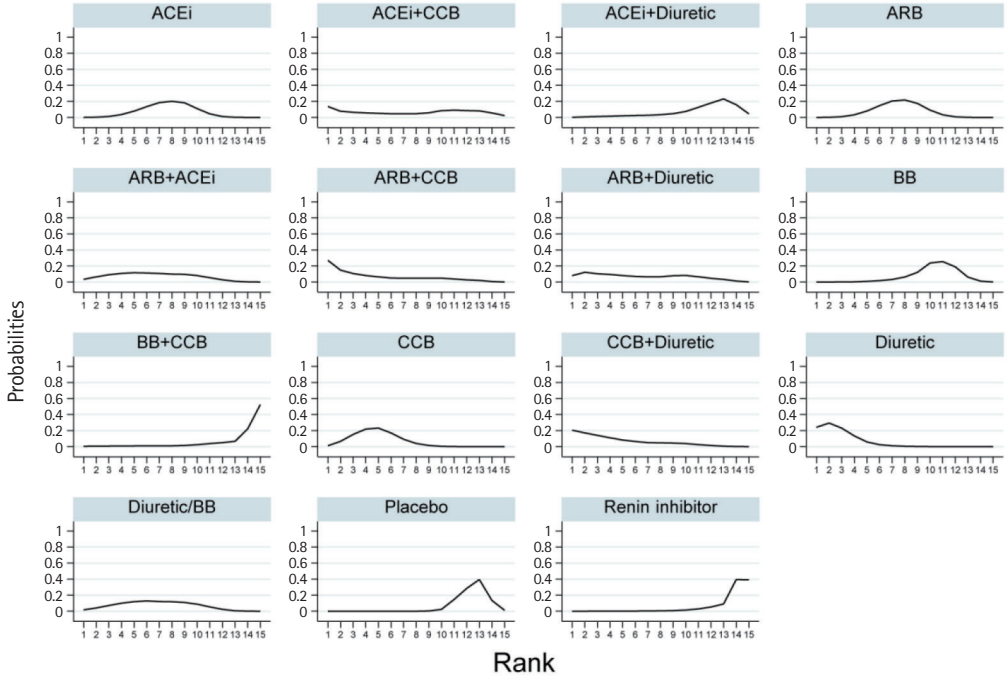


Supplementary Figure 27. Sensitivity analysis for all cause death. OR, odds ratio; CI, confidence interval.

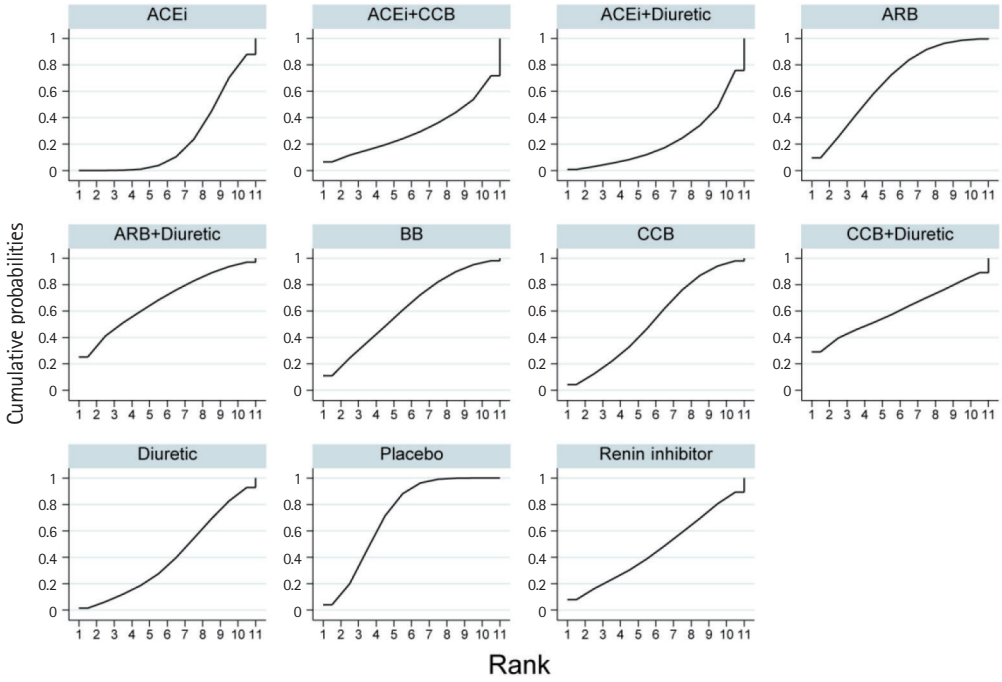
https://doi.org/10.5853/jos.2020.02698



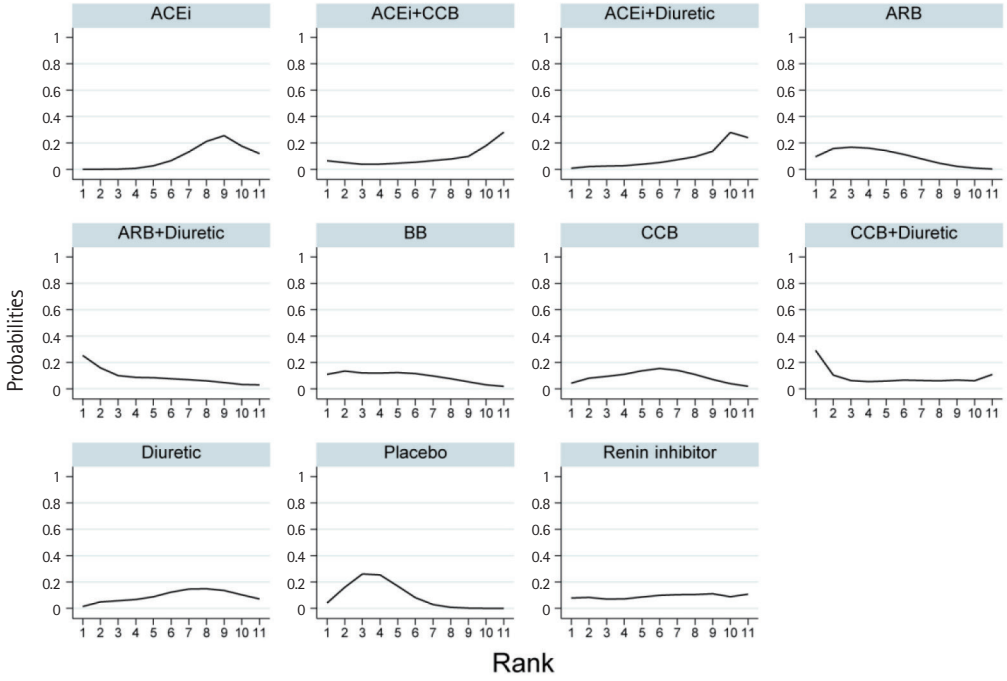
Supplementary Figure 28. The surface under the cumulative ranking curve (SUCRA) values of multiple treatments for efficacy. SUCRA, ranging from 1 to 0, indicating that the treatment has a high likelihood of being best and has a high likelihood of being worst, respectively. ACEi, angiotensin-converting enzyme inhibitor; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; BB, beta-blocker.



Supplementary Figure 29. Ranking of multiple treatments for efficacy. Ranking positions for all interventions (1 [best] to 15 [worst]). ACEi, angiotensin-converting enzyme inhibitor; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; BB, beta-blocker.

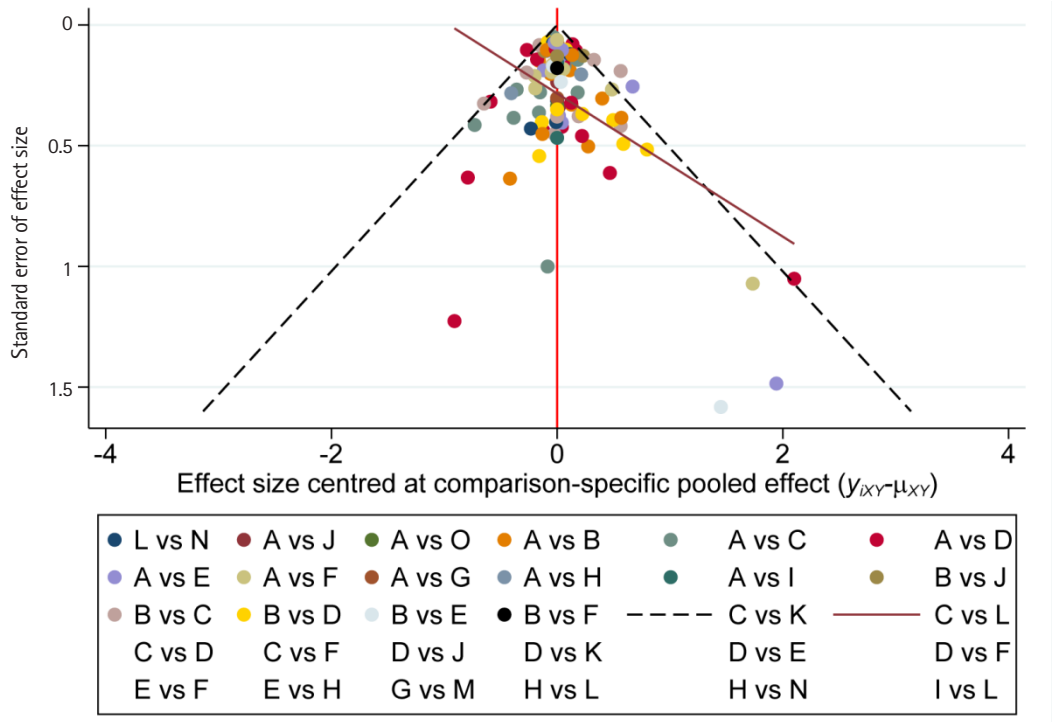


Supplementary Figure 30. The surface under the cumulative ranking curve (SUCRA) values of multiple treatments for tolerability. SUCRA, ranging from 1 to 0, indicating that the treatment has a high likelihood of being best and has a high likelihood of being worst, respectively. ACEi, angiotensin-converting enzyme inhibitor; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; BB, beta-blocker.

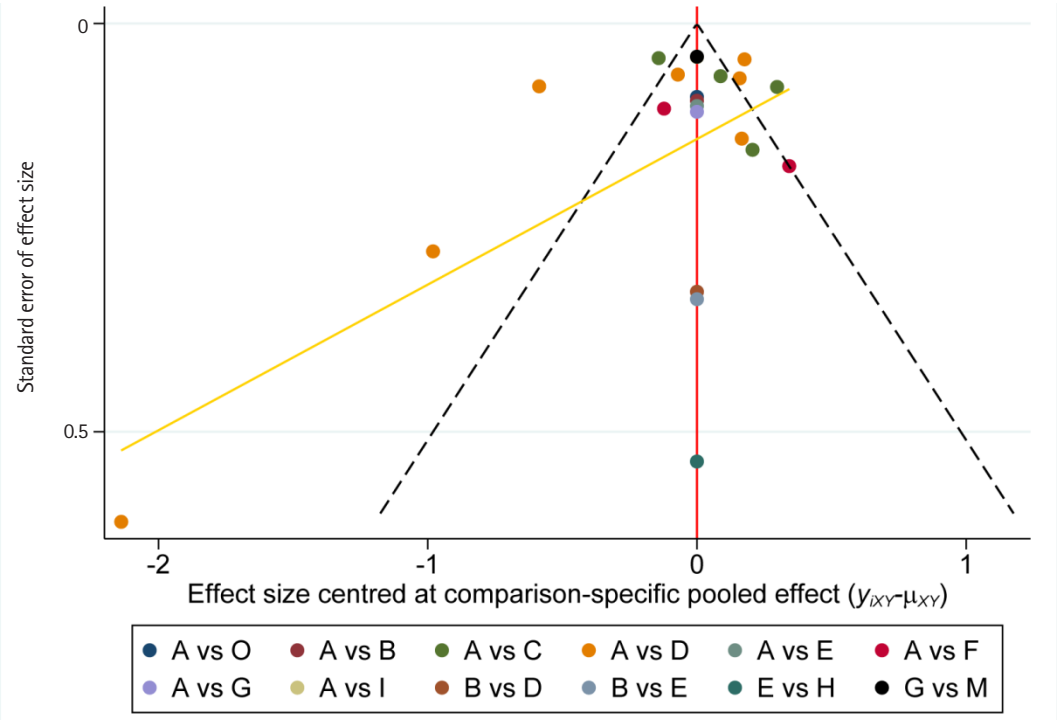


Supplementary Figure 31. Ranking of multiple treatments for tolerability. Ranking positions for all interventions (1 [best] to 15 [worst]). ACEi, angiotensin-converting enzyme inhibitor; CCB, calcium channel blocker; ARB, angiotensin II receptor blocker; BB, beta-blocker.

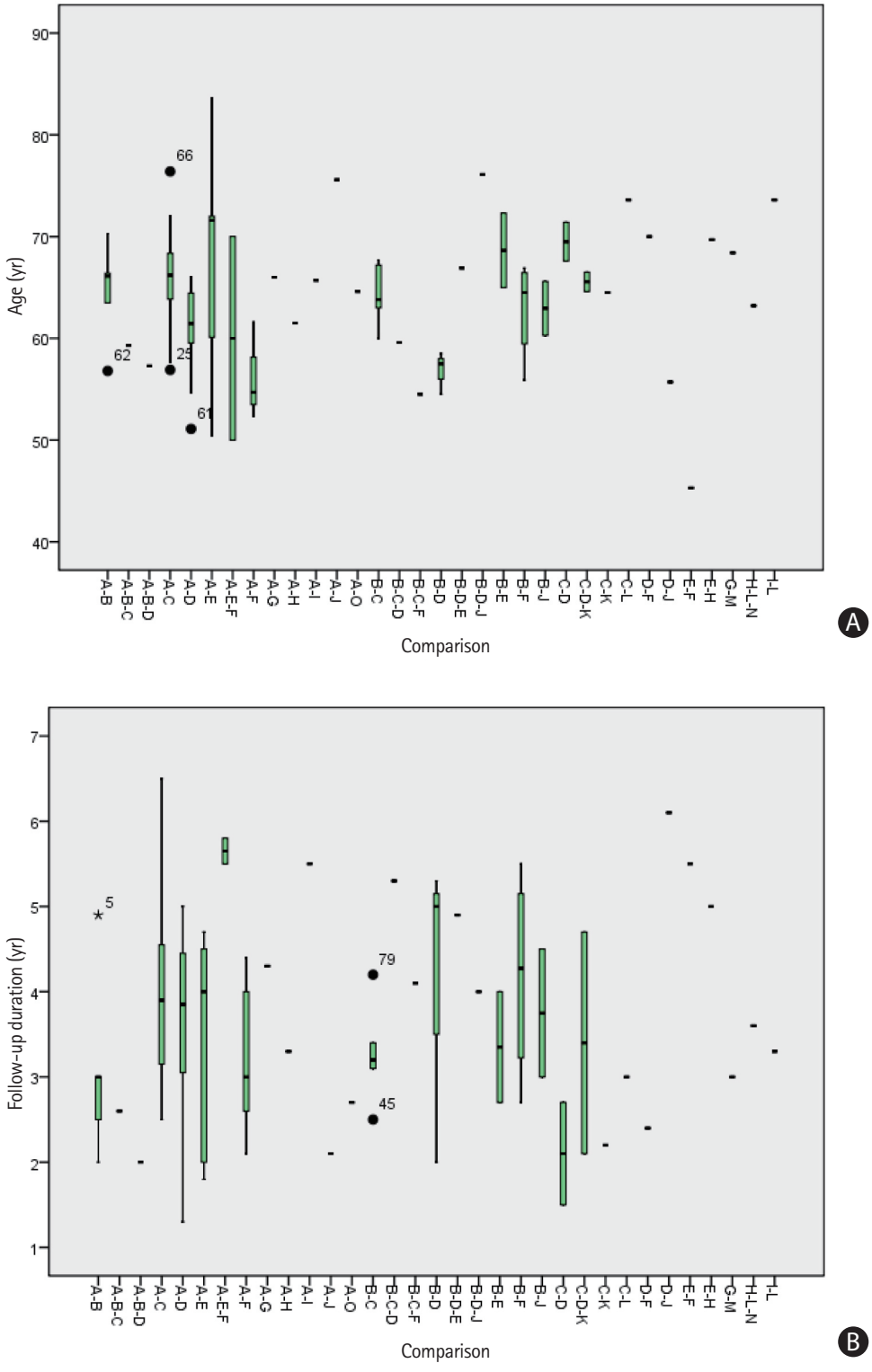
https://doi.org/10.5853/jos.2020.02698



Supplementary Figure 32. Comparison-adjusted funnel plot for efficacy. A, placebo; B, calcium channel blocker (CCB); C, angiotensin II receptor blocker (ARB); D, angiotensin-converting enzyme inhibitor (ACEi); E, diuretic; F, beta-blocker (BB); G, ACEi+diuretic; H, CCB+diuretic; I, ARB+diuretic; J, BB/diuretic; K, ARB+ACEi; L, ARB+CCB; M, ACEi+CCB; N, BB+CCB; O, renin inhibitor. This is drawn only for comparisons with two or more studies.



Supplementary Figure 33. Comparison-adjusted funnel plot for tolerability. A, placebo; B, calcium channel blocker (CCB); C, angiotensin II receptor blocker (ARB); D, angiotensin-converting enzyme inhibitor (ACEi); E, diuretic; F, beta-blocker (BB); G, ACEi+diuretic; H, CCB+diuretic; I, ARB+diuretic; M, ACEi+CCB; O, renin inhibitor. This is drawn only for comparisons with two or more studies.



Supplementary Figure 34. Assessment of transitivity. (A) Age. (B) Follow-up duration. A, placebo; B, calcium channel blocker (CCB); C, angiotensin II receptor blocker (ARB); D, angiotensin-converting enzyme inhibitor (ACEi); E, diuretic; F, beta-blocker (BB); G, ACEi+diuretic; H, CCB+diuretic; I, ARB+diuretic; J, diuretic/BB; K, ARB+ACEi; L, ARB+CCB; M, ACEi+CCB; N, BB+CCB; O, renin inhibitor.

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